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Department : Mathématiques et Informatiques (MI)

Educational handout

Algebra 1: Lesson and Solved Exercises

For the benefit of students "1st year- Computer Science (L1)"

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ALGBRA

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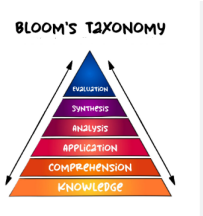
General Objectives

By the end of this course, students will be able to:



- **Explain** the basic concepts of formal logic used in mathematical reasoning.
- **Apply** set operations and functions to algebraic and computational problems.
- **Analyze** binary relations on a set using standard mathematical properties.
- **Construct** proofs involving algebraic structures such as groups, rings, and fields.
- **Examine** polynomial rings through algebraic operations and structural properties.
- **Evaluate** algebraic models in relation to applications in computer science.

Prerequesties

To succeed in this Algebra 1 course, its helpful to have the following basics:

- Basic Arithmetic Skills: Knowing how to add, subtract, multiply, and divide, as well as understanding fractions and percentages.
- Familiarity with Algebra Concepts: Having some knowledge of variables, expressions, and solving simple equations.
- Problem-Solving Ability: Being able to logically manipulate equations and solve problems step by step.

Introduction

This module is designed for first-year computer science students(L1) at Abdelhamid Mehri University. It provides a solid foundation in fundamental algebraic concepts, covering a range of topics crucial for advanced studies in computer science. The main chapters of the course include:

Logic and Proofs: Introduction to logical reasoning, truth tables, and various proof techniques, essential for developing mathematical rigor.

Sets and Functions: Exploration of sets, functions, and their properties, with applications in different mathematical structures.

Binary Relations on a Set: Study of binary relations on a set, including equivalence relations and order relations.

Algebraic Structures: Introduction to key algebraic structures such as groups, rings, and fields.

Polynomial Rings: Study of polynomials and their degree, construction of the polynomial ring, polynomial arithmetic and the study of polynomial roots.

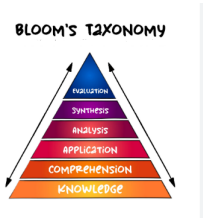
This module not only strengthens problem-solving skills but also provides essential tools for applications in computer science, such as data structures, algorithms, and the formalization of computational processes.

1

Basic Concepts of Logic

Specific Objectives

By the end of this chapter, students will be able to:



1. Define fundamental concepts such as propositions, logical connectives, truth values, and quantifiers.
2. Explain truth tables of compound propositions that are expressed using logical connectives.
3. Translate everyday language statements into symbolic form employing quantifiers appropriately.
4. Simplify compound propositions using properties of the logical connectives.
5. Apply logical reasoning and proof techniques in computer science contexts.
6. Develop mathematical proofs using direct proof, proof by contrapositive, proof by contradiction, and mathematical induction.

Introduction

The basic concepts of logic form the foundation of reasoning in computer science. Propositional logic provides the formal language needed to express propositions precisely. Studying logic is particularly useful in computer science, not only because it leads to Boolean logic, the basis of digital circuits and binary computation, but also because it develops the ability to reason formally, analyze algorithms, and verify program correctness. Methods of proof supply the rigorous techniques required to establish correctness. Together, these tools are essential in areas such as algorithms, program verification, databases, and digital systems. Moreover, understanding logical reasoning and proof techniques is also fundamental for mastering other core subjects studied by computer science students, such as analysis and linear algebra, where

precise reasoning and formal proofs play a central role. This chapter is divided into two sections: the first covers propositional logic, while the second focuses on methods of proof.

1.1 Truth Tables

In this section, we introduce the definition of a logicalpropositionand the logical connectives which are a formal language for expressing propositions and reasoning about their truth using **truth tables**.

1.1.1 Propositions

Definition

A proposition is a statement (declarative sentence) for which one can determine, without ambiguity, whether it is true or false, but never both at the same time.

Definition

A truth value, sometimes called a logical value, is a value indicating the relation of a proposition to truth, which has only two possible values: True (T or 1) and False (F or 0).

Examples

Declarative sentences that are propositions:

- "Every rectangle is a square." (A declarative sentence which is **false**, so it is a proposition.)
- "2+4=6." (A declarative sentence which is **true**, so it is a proposition.)
- "The sky is blue." (A declarative sentence which is **true**, so it is a proposition.)
- "For all $z \in \mathbb{C}$, we have $|z| = 2$." (A declarative sentence which is **false**, so it is a proposition.)

Sentences that are not propositions:

- "Close the door!" (This is a command, not a declarative sentence.)
- "What time is it?" (This is a question, not a declarative sentence.)
- $x + 2 > 5$ (This is an open mathematical sentence: it has no truth value until we know the value of x.)
- n is a prime number (This is not a proposition unless a specific value of n is given.)

Remark

- Every proposition is a declarative sentence, but not every declarative sentence is a proposition.
- Interrogative, imperative and exclamatory sentences are not propositions because they are not declarative sentences and do not have a truth value.
- Open mathematical sentences (with variables) are not propositions until the variables are specified(with a value or in a set).

Definition

A compound proposition is a new proposition formed from two or more simple propositions using logical connectives.

Definition

A truth table is a table that displays the truth value of a proposition (or of a compound proposition) for all possible truth values of its components.

Remark

There are n columns for the simple propositions $P_1, P_2, \ldots, P_n$. The table contains 2^n rows showing all possible truth values of the n propositions, and the last column gives the truth value of the compound proposition for each possibility.

Let P and Q be propositions. We shall now define new propositions built from P and Q using logical connectives.

1.1.2 Logical Connectives

In this part, we give the main logical connectives used to form compound propositions and their corresponding truth tables.

a. Negation

Definition

The negation of a proposition P is a new proposition whose truth value is the opposite of that of P. The notation for the negation of P, is

$$\neg P \text{ or } \overline{P},$$

and it is read as "Not P" or "It is not the case that P". The proposition $\overline{P}$ is true when P is false and false when P is true.

Truth Table for Negation

P	$\overline{P}$
T	F
F	T

Examples (Negation).

Let P: "Algiers is the capital of Algeria." Here, P is **true**, so the negation $\overline{P}$: "Algiers is not the capital of Algeria" is **false**.

Now, let P: "10 is a prime number." Here, P is **false**, so the negation $\overline{P}$: "10 is not a prime number" is **true**.

b. Conjunction

Definition

The conjunction of two propositions P and Q is a new proposition formed by combining P and Q using the logical connective called "conjunction" noted $\wedge$. This new proposition is denoted by

$$P \wedge Q,$$

and is read as "P and Q."
The proposition $P \wedge Q$ is true only when both P and Q are true. In every other case (if one is false or both are false), $P \wedge Q$ is false.

Truth Table for Conjunction

P	Q	$P \wedge Q$
T	T	T
T	F	F
F	T	F
F	F	F

Examples (Conjunction)

1. P : "The square root of 9 is 3" (true). Q : "5 is divisible by 2" (false). Then $P \wedge Q$: "The square root of 9 is 3 and 5 is divisible by 2" is **false**.
2. P : "The square root of 9 is 3" (true). Q : "4 is divisible by 2" (true). Then $P \wedge Q$: "The square root of 9 is 3 and 4 is divisible by 2" is **true**.
3. P : "The square root of 12 is 3" (false). Q : "4 is divisible by 2" (true). Then $P \wedge Q$: "The square root of 12 is 3 and 4 is divisible by 2" is **false**.
4. P : "The square root of 12 is 3" (false). Q : "5 is divisible by 2" (false). Then $P \wedge Q$: "The square root of 12 is 3 and 5 is divisible by 2" is **false**.

c. Disjunction

Definition

The disjunction of two propositions P and Q is a new proposition composed by combining two propositions P and Q using the logical connective "disjunction" noted $\vee$. This new proposition is denoted by

$$P \vee Q,$$

and is read as "P or Q."
The proposition $P \vee Q$ is true if at least one of P or Q is true, and it is false only when both P and Q are false.

Truth Table for Disjunction

P	Q	$P \vee Q$
T	T	T
T	F	T
F	T	T
F	F	F

Examples (Disjunction).

1. P : "7 is a prime number" (true). Q : "10 is odd" (false). Then $P \vee Q$: "7 is prime or 10 is odd" is **true**.
2. P : "8 is a prime number" (false). Q : "10 is odd" (false). Then $P \vee Q$: "8 is prime or 10 is odd" is **false**.
3. P : "7 is a prime number" (true). Q : "9 is odd" (true). Then $P \vee Q$: "7 is prime or 9 is odd" is **true**.
4. P : "8 is a prime number" (false). Q : "9 is odd" (true). Then $P \vee Q$: "7 is prime or 10 is odd" is **true**.

d. Implication

Definition

The implication between two propositions P and Q is the proposition "$\overline{P} \vee Q$" noted $P \Rightarrow Q$ and it can be read as "if P, then Q". It is only false when P is true and Q is false; otherwise it is true. In an implication, P is called the hypothesis or antecedent , and Q is called the conclusion or consequence. The proposition $P \Rightarrow Q$ is always true if P is false. It can be read in several equivalent ways:

- "P implies Q"
- "If P, then Q"
- "Q whenever P"
- "Q follows from P"
- "P is sufficient for Q"
- "Q is necessary for P"

The proposition $P \Rightarrow Q$ is also called conditional proposition.

Truth table of the implication:

P	Q	$\overline{P} \vee Q$	$P \Rightarrow Q$
T	T	T	T
T	F	F	F
F	T	T	T
F	F	T	T

Examples (Implication)

- Let P: "$\sqrt{2} > 0$" and Q: "$1 > 0$". The implication $P \Rightarrow Q$ is **true** (dividing by $\sqrt{2}$).
- Let P: "$\sqrt{2} > 0$" and Q: "$1 < 0$". The implication $P \Rightarrow Q$ is **false**
- Let P: "$2 + 2 = 5$" and Q: "$\sqrt{2} = 2$". The implication $P \Rightarrow Q$ is **true**, because when P is false, the implication is always true.
- Let P: "$2 + 2 = 5$" and Q: "$\sqrt{4} = 2$". The implication $P \Rightarrow Q$ is **true**, because when P is false, the implication is always true.

e. Biconditional

Definition

Let P and Q be two propositions. A biconditional is a logical connective that represents a relationship between two propositions P and Q. The biconditional proposition denoted

$$P \Leftrightarrow Q,$$

and defined as

$$P \Leftrightarrow Q \text{ is } (P \Rightarrow Q) \wedge (Q \Rightarrow P).$$

This notation is read as:

- "P is equivalent to Q" ;
- "P if and only if Q" (often abbreviated "iff").

Remark

The biconditional proposition is true when P and Q are both true, or when P and Q are both false.

Truth table of the Biconditional:

P	Q	$P \Rightarrow Q$	$Q \Rightarrow P$	$(P \Rightarrow Q) \wedge (Q \Rightarrow P)$	$P \Leftrightarrow Q$
T	T	T	T	T	T
T	F	F	T	F	F
F	T	T	F	F	F
F	F	T	T	T	T

Examples (Biconditional)

1. We have the following biconditional proposition: For $x, x' \in \mathbb{R}$,

$$x \cdot x' = 0 \quad \Leftrightarrow \quad (x = 0 \ \text{ or } \ x' = 0)$$

is true.

2. Here is an equivalence that is always false (for any proposition P):

$$P \Leftrightarrow \overline{P}.$$

Properties of Logical Connectives

Let P, Q, R be propositions. We have the following properties:

1. $P \iff \overline{\overline{P}}$
2. $P \wedge Q \iff Q \wedge P$ (commutativity of $\wedge$)
3. $P \vee Q \iff Q \vee P$ (commutativity of $\vee$)
4. $(P \wedge Q) \wedge R \iff P \wedge (Q \wedge R)$ (associativity of $\wedge$)
5. $(P \vee Q) \vee R \iff P \vee (Q \vee R)$ (associativity of $\vee$)
6. $P \wedge (Q \vee R) \iff (P \wedge Q) \vee (P \wedge R)$ (distributivity of $\wedge$ over $\vee$)
7. $P \vee (Q \wedge R) \iff (P \vee Q) \wedge (P \vee R)$ (distributivity of $\vee$ over $\wedge$)
8. $\overline{P \wedge Q} \iff \overline{P} \vee \overline{Q}$ (De Morgan's law)
9. $\overline{P \vee Q} \iff \overline{P} \wedge \overline{Q}$ (De Morgan's law)
10. $\overline{P \Rightarrow Q} \iff P \wedge \overline{Q}$ (negation of implication)
11. $(P \Rightarrow Q) \iff (\overline{Q} \Rightarrow \overline{P})$ (contrapositive)
12. $(P \Rightarrow Q) \wedge (Q \Rightarrow R) \Rightarrow (P \Rightarrow R)$ (transitivity)
13. The converse of $(P \Rightarrow Q)$ is $(Q \Rightarrow P)$

1.1.3 Additional Logical Notions

Definition

A **tautology** is a proposition that is always true, no matter the truth values of its components. For example, the proposition $P \vee \overline{P}$ is a tautology. Its truth table is:

P	$\overline{P}$	$P \vee \overline{P}$
T	F	T
F	T	T

Definition

A **contradiction** is a proposition that is always false, no matter the truth values of its components. For example, the proposition $P \wedge \overline{P}$ is a contradiction. Its truth table is:

P	$\overline{P}$	$P \wedge \overline{P}$
T	F	F
F	T	F

Definition

Let P and Q be two propositions. The **exclusive or** of P and Q, denoted by $P \underline{\vee} Q$, is the proposition that is true when exactly one of P and Q is true and is false otherwise. The truth table of $P \underline{\vee} Q$:

P	Q	$P \underline{\vee} Q$
T	T	F
T	F	T
F	T	T
F	F	F

1.2 Quantifiers

We introduce quantifiers, which allow us to formulate precise mathematical propositions about all or some elements of a set. We first introduce the definition of a predicate, and then we present the definitions of quantifiers.

Definition

A predicate is a mathematical expression that describes a property of one or more variables. By itself, it is not true or false. When we give values to the variables, it becomes a proposition (true or false).

Examples

Predicate: $P(x) : x > 5$

- If $x = 7$, then $P(7)$ is true.
- If $x = 3$, then $P(3)$ is false.

1.2.1 The Universal Quantifier $\forall$

Definition

The proposition

$$\forall x \in E : \ P(x)$$

is **true** when the predicate $P(x)$ is true for every element x in the set E.
It is read as: "For all x belonging to E, $P(x)$", meaning "For all x in E, the predicate $P(x)$ is true."

Examples:

- $\forall x \in [1, +\infty[: \ (x^2 > 1)$ is true.
- $\forall x \in \mathbb{R} : \ (x^2 > 1)$ is false.

1.2.2 The Existential Quantifier $\exists$

Definition

The proposition

$$\exists x \in E : \ P(x)$$

is **true** when there is at least one element x in E for which $P(x)$ is true.
It is read as: "There exists x in E such that $P(x)$."

Examples:

- $\exists x \in \mathbb{R} : \ (x(x-1) < 0)$ is true (We can take $x = \frac{1}{2}$).
- $\exists n \in \mathbb{N} : \ n^2 - n > n$ is true (We can take $n = 3$).
- $\exists x \in \mathbb{R} : \ (x^2 = -1)$ is false (since no real number squared is negative).

Remark

The existential quantifier with uniqueness denoted by the symbol $\exists!$ which means "there exists a unique". The statement

$$\exists!\, x \in E : \; P(x)$$

is true if there is **exactly one** element x in E such that $P(x)$ is true. For example

$$\exists!\, n \in \mathbb{N} : \; n - 2 = 0$$

is true, because the equation $n - 2 = 0$ has exactly one solution in $\mathbb{N}$, namely $n = 2$.

1.2.3 Negation of Quantifiers

- The negation of $\forall x \in E : \; P(x)$ is $\exists x \in E : \; \overline{P(x)}$.

  Examples:
  The negation of $\forall x \in [1, +\infty[: \; (x^2 > 1)$ is $\exists x \in [1, +\infty[: \; (x^2 \leq 1)$.
  The negation of $\forall x \in \mathbb{R} : \; (x + 1 \in \mathbb{Z})$ is $\exists x \in \mathbb{R} : \; (x + 1 \notin \mathbb{Z})$.

- The negation of $\exists x \in E : \; P(x)$ is $\forall x \in E : \; \overline{P(x)}$.

  Example:
  The negation of $\exists z \in \mathbb{C} : \; (z^2 + z + 1 = 0)$ is $\forall z \in \mathbb{C} : \; (z^2 + z + 1 \neq 0)$.

- For complex propositions: The negation of

$$\forall x \in E, \; \exists x \in E : \; P(x).$$

  is

$$\exists x \in E, \; \forall x \in E : \; \overline{P(x)}.$$

  Example:
  The negation of

$$\forall x \in \mathbb{R}, \; \exists y > 0 : \; (x + y > 10)$$

  is

$$\exists x \in \mathbb{R}, \; \forall y > 0 : \; (x + y \leq 10).$$

1.2.4 Importance of the Order of Quantifiers

The order of quantifiers is very important. For example:

1. $\forall x \in \mathbb{R}, \exists y \in \mathbb{R} : \; x > y$ is true. (For each x, we choose $y = x - 1$.)
2. $\exists x \in \mathbb{R}, \forall y \in \mathbb{R} : \; x > y$ is false. (The same x cannot be greater than all real numbers y.)

This is similar to natural language:

- True: "For every person, there exists a phone number" (the number depends on the person).
- False: "There exists a phone number for all people" (the same number for everyone).

1.3 Methods of Proof

In mathematics, proofs provide the logical foundation for establishing the truth of propositions. In this section, we present some fundamental methods of proof.

1.3.1 Direct Proof

Definition

A **direct proof** shows that a conditional proposition $P \Rightarrow Q$ is true by assuming P is true and then deducing Q using axioms, definitions, theorems and previously proven results. Thus, the situation "P true and Q false" never occurs.

Example:
We prove that: For all $n \in \mathbb{Z}$: if n is odd, then n^2 is odd.
We suppose that n is an odd integer, then $\exists k \in \mathbb{Z} :\ n = 2k + 1$. Hence

$$n^2 = (2k+1)^2 = 4k^2 + 1 + 4k = 2(2k^2 + 2k) + 1 = 2k' + 1, \text{ with } k' = 2k^2 + 2k,\ k' \in \mathbb{Z},$$

hence, n^2 is odd.

1.3.2 Proof by Contradiction

Definition

In a proof by contradiction, to prove that P is true, we assume the negation of P i.e. $\overline{P}$. If we can find a contradiction Q (for example, a proposition of the form $R \wedge \overline{R}$ with R is a proposition) such that $\overline{P} \Rightarrow Q$ is true. Because Q is false, but $\overline{P} \Rightarrow Q$ is true, we can deduce that $\overline{P}$ is false, which means that P is true.

Example:
Prove that

$$P: \ \forall a, b \in \mathbb{R}^+ : \left(\frac{a}{1+b} = \frac{b}{1+a} \quad \Longrightarrow \quad a = b \right).$$

We will prove this by contradiction. We assume the negation of the proposition P that

$$\overline{P}: \ \exists a, b \in \mathbb{R}^+ : \left(\frac{a}{1+b} = \frac{b}{1+a} \ \wedge \ a \neq b \right).$$

From the equality we obtain:

$$a(1+a) = b(1+b).$$

This simplifies to

$$a + a^2 = b + b^2,$$

then,

$$a^2 - b^2 = b - a.$$

Hence,

$$(a-b)(a+b) = -(a-b).$$

Since $a \neq b$, it follows that $a - b \neq 0$. Dividing both sides by $(a-b)$ gives

$$a + b = -1.$$

But we have $a, b \in \mathbb{R}^+$, which implies $a + b > 0$. Hence we obtain

$$a + b = -1 \text{ and } a + b > 0,$$

which is a contradiction. Therefore, our hypothesis $\overline{P}$ leads to a contradiction. Hence $\overline{P}$ is false then P is true.

1.3.3 Proof by Contrapositive

Definition

A conditional proposition $P \Rightarrow Q$ is logically equivalent to its contrapositive $\overline{Q} \Rightarrow \overline{P}$. In a proof by contrapositive, we prove $\overline{Q} \Rightarrow \overline{P}$ instead of $P \Rightarrow Q$. We suppose that $\overline{Q}$ is true and we prove that $\overline{P}$ is true.
This is useful when proving P implies Q is difficult, but assuming $\overline{Q}$ allows us to deduce $\overline{P}$.

Example:
We prove that: If n^2 is even, then n is even.
Its contrapositive is: If n is odd, then n^2 is odd. From the example of the direct proof this contrapositive is already proved then it is true, and so the original proposition is true.

1.3.4 Proof by Induction

Definition

A proof by induction is used to show that a predicate $P(n)$ holds for all natural numbers $n \geq n_0$. It consists of two steps:

1. **Base case:** Prove that $P(n_0)$ is true.
2. **Inductive step:** Assume $P(n)$ is true for some $n \in \mathbb{N}$, $n \geq n_0$, and prove that $P(n+1)$ is true.

If both steps hold, then by the principle of mathematical induction the proposition "$P(n)$ is true for all natural numbers $n \geq n_0$" is true.

Example:
For all natural number $n \geq 1$,

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}.$$

Proof:
Base case: For $n = 1$, $P(1)$ is true.
Inductive step: We suppose that $P(n)$ is true for some natural number $n \geq 1$ and we prove $P(n+1)$ is true. We assume that the proposition holds for $n = k$, i.e.,

$$1 + 2 + \cdots + k = \frac{k(k+1)}{2}.$$

Then by adding $k+1$ to both sides, we get

$$1 + 2 + \cdots + k + (k+1) = \frac{k(k+1)}{2} + (k+1),$$

we have:

$$\frac{k(k+1) + 2(k+1)}{2} = \frac{(k+1)(k+2)}{2} = \frac{(k+1)((k+1)+1)}{2},$$

thus,

$$1 + 2 + \cdots + k + (k+1) = \frac{(k+1)((k+1)+1)}{2}.$$

Therefore, the proposition holds for $k+1$.
In conclusion, from the two previous steps and by the principle of mathematical induction, for all $n \geq 1$,

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}.$$

We have the following proofs that are also useful in mathematical reasoning.

Proof by Counter-example

Definition

A proof by counter-example is a reasoning method used to show that a general proposition is false by finding a single particular case, or counter-example, where the proposition does not hold. To refute a universal proposition "for all $x \in E$: $P(x)$ ", it is enough to find an x for which P is false.

Example:
All squares of integers are divisible by 3.
Counterexample: $2^2 = 4$, which is not divisible by 3. Hence, the proposition is false.

Proof by Cases

Definition

A proof by cases is a method used to show that a predicate $P(x)$ holds for all x in a set E by dividing E into parts (subsets) and proving $P(x)$ for each part. If the predicate holds in all parts, then the proposition "$P(x)$ for all $x \in E$" is true.

Example:
We prove that for any integer n, $n^2 + n$ is even.

Proof:

- Case 1: If $\exists k \in \mathbb{Z} : n = 2k$ (n is even). Then $n^2 + n = 4k^2 + 2k = 2(2k^2 + k) = 2k'$, with $k' \in \mathbb{Z}$, thus $n^2 + n$ is even.
- Case 2: If $\exists k \in \mathbb{Z} : n = 2k + 1$ (n is odd). Then $n^2 + n = (2k + 1)^2 + (2k + 1) = 2(2k^2 + 3k + 1) = 2k''$, with $k'' \in \mathbb{Z}$, thus $n^2 + n$ is even.

Hence, $\forall n \in \mathbb{Z} : n^2 + n$ is even.

Test Your Chapter Mastery

Exercise 1: Compound Propositions

Let P, Q and R be propositions

1. If P is false , Q is true and R is false . What are the true propositions?
   (1) $Q \wedge (P \vee R)$ (2) $P \vee (Q \wedge R)$ (3) $(P \vee Q) \wedge (Q \vee R)$ (4) $\overline{(P \wedge Q \wedge R)}$

2. Evaluate the following propositions by considering only the truth values of the propositions that are given:
   1. $Q \Rightarrow (P \Rightarrow R)$, if Q is false 2. $P \wedge (P \vee Q)$, if Q is true.
   3. $P \vee (Q \Rightarrow R)$, if Q is false 4. $(Q \wedge P) \wedge (\overline{P} \vee R \vee \overline{Q})$ if P is false

3. i. Which of the following propositions are tautologies?
   $P_1 : (P \vee Q) \implies Q \quad P_2 : (P \vee \overline{Q}) \vee (\overline{P} \wedge Q) \quad P_3 : (P \implies Q) \vee (Q \implies P)$
   $P_4 : P \wedge (Q \underline{\vee} R) \Leftrightarrow (P \wedge Q) \underline{\vee} (P \wedge R)$
   ii. Give the negation of P_1, P_2 and P_3.

4. i. Using the truth table prove the following biconditional propositions:
   $(\overline{P \wedge Q}) \Leftrightarrow \overline{P} \vee \overline{Q}, \quad (\overline{P \vee Q}) \Leftrightarrow \overline{P} \wedge \overline{Q}, \quad P \Rightarrow Q \Leftrightarrow \overline{Q} \Rightarrow \overline{P}$
   ii. Deduce that the previous biconditional propositions are tautologies.

5. Compare the propositions in the two following cases:
   1) $P \vee Q$ with $P \vee (Q \Rightarrow R)$. 2) $P \vee P$ with P

Exercise 2: From Text to Quantifiers

Let f be a function defined from $\mathbb{R}$ to $\mathbb{R}$. Express the following statements using quantifiers:

1. f is not constant.
2. f is decreasing.
3. f is an even function.
4. f is bounded.

Exercise 3: From Quantifiers to Text

Let f be a function defined from $I \subset \mathbb{R}$ to $\mathbb{R}$. Write in words the meaning of the following statements:

1. $\exists A \in \mathbb{R}, \forall x \in I \subseteq \mathbb{R} \,/ f(x) = A$.
2. $\exists t \in \mathbb{R}^*, \forall x \in I \subseteq \mathbb{R} \,/ (x + t \in I \wedge f(x + t) = f(x))$.
3. $\forall x_1, x_2 \in I \subseteq \mathbb{R} \,/ (x_1 > x_2 \Rightarrow f(x_1) > f(x_2))$.
4. $\forall x_1, x_2 \in I \subseteq \mathbb{R} \,/ (x_1 \neq x_2 \Rightarrow f(x_1) \neq f(x_2))$.

Exercise 4: Truth value and Negation

Let the following propositions:

- $\exists x \in \mathbb{R}, \forall y \in \mathbb{R} \mid x + y \geq 0$.
- $\forall x \in \mathbb{R}, \exists y \in \mathbb{R} \mid x + y \geq 0$.
- $\forall n \in \mathbb{N}, \exists k \in \mathbb{N} \mid n = k^2 + 1$.
- $\forall n \in \mathbb{N}, \exists k \in \mathbb{N} \mid n = 2k + 1$.
- $\exists x \in \mathbb{R}, \forall y \in \mathbb{R} \mid y^2 > x$.

- $\forall k \in \mathbb{Z} \mid -2 < k < 4$.

1. Give the negation of each proposition.
2. These propositions are true or false ? Justify.

Exercise 5: Recurrence

Prove the following proposition:

$$\forall n \in \mathbb{N},\ \forall x > 0 :\ (1+x)^n \geq 1 + nx.$$

Exercise 6: Rational and Irrational

1. Prove that $\sqrt{2}$ is irrational i.e. $\sqrt{2} \notin \mathbb{Q}$.
2. Prove that $\sqrt{4-2\sqrt{3}} - \sqrt{4+2\sqrt{3}}$ is a rational number.

Exercise 7: Divisiblity by 8

We want to prove the following property, for $n \in \mathbb{N}^*$:

$$\text{If } (n^2-1) \text{ is not divisible by } 8,\ \text{ then } n \text{ is even.}$$

1. Write the contrapositive of the previous statement.

   Noticing that for $n \in \mathbb{N}^*$, n is odd can be written in the form

   $$n = 4k + r \quad \text{with } k \in \mathbb{N} \text{ and } r \in \{1, 3\} \quad \text{(to be justified)},$$

2. Prove the contrapositive.
3. Have we proved the property of the statement?

Exercise 8: False Proposition

Let P be the proposition:

$$\exists a \in]-\infty, 0[, \forall x \in [-1, +\infty[:\ \Big(x > a \implies \Big[(x^2 > \tfrac{a^2}{4})\ \text{ or }\ (x \leq 0)\Big]\Big).$$

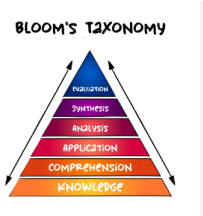
- Prove that P is false.

2

Sets and functions

Specific Objectives

By the end of this chapter, students will be able to:



1. Define basic set concepts, notations, and types of sets.
2. Describe set operations and their relationships using Venn diagrams.
3. Determine subsets, power sets, Cartesian products, and set cardinalities.
4. Explain the concept of a function and its notation and its relation with sets.
5. Classify functions as injective, surjective, or bijective.
6. Apply function composition and inverse functions.

Introduction

Sets and functions are among the fundamental concepts in mathematics. They form the foundation for mastering other core subjects such as analysis and linear algebra, which are essential for computer science students. Since sets and functions are closely related, we begin by defining sets, presenting their main properties, and describing the basic operations on sets. Then, we introduce functions as relations between sets, along with their essential properties. This chapter is divided into two sections: the first focuses on sets, and the second on functions.

2.1 Sets: Definitions and Examples

Mathematics deals with objects of very different kinds; from your previous experience, you are familiar with many of them: Numbers, points, lines, planes, triangles, circles, angles, equations, functions and many more. Often, objects of a similar nature or with a common property are collected into sets; these may be finite or infinite.

In this section, we will define sets, learn how to represent them, and study the main operations between sets such as union, intersection, and difference along with their fundamental properties.

Definition

A set is a well-defined collection of distinct objects. The objects in a set are called the elements of the set. A set is said to contain its elements. The notation $a \in A$ denotes that a is an element of the set A, i.e. a belongs to A. If a is not a member of A, i.e. a does not belong to the set A, we write $a \notin A$.

Example

If G is the set of all even numbers, then 16 is an element of G and we write $16 \in G$, and 3 does not belong to G and we write $3 \notin G$.

Remark

- Set is denoted by capital letters $A, B, C...$
- Elements of set are denoted by small letters $a, b, c...$

Finite and Infinite Sets

Definition

A set A is called finite, if it has no elements, or if there is a natural number n such that the elements of A can be numbered $1, ..., n$ in such a way that every element of A appears exactly once. Otherwise, A is called infinite.

Definition

Let A be a finite set, the number of different elements in the set A is called its cardinality and it is denoted by $card(A)$ or $|A|$.

Example

$A = \{1, 2, 3, 4, 5\}$ is a finite set because it has 5 elements, then $|A| = 5$.
$\mathbb{N} = \{1, 2, 3, 4, \ldots\}$ is infinite since it continues endlessly.

We also introduce the usual sets of numbers

$$\mathbb{N} = \{0, 1, 2, 3, \ldots\}, \quad \mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}, \quad \mathbb{Q} = \left\{\frac{p}{q} \mid p, q \in \mathbb{Z},\ q \neq 0\right\},$$

$$\mathbb{R} = \{x \text{ is a real number}\}, \quad \mathbb{C} = \{a + bi \mid a, b \in \mathbb{R},\ i^2 = -1\}.$$

2.1.1 Representations of Sets

In this part, we give the different ways to represent a set.

1. Roster or Tabular Form

In this form, all elements of the set are listed explicitly between braces (curly brackets) $\{\}$, separated by commas. For example:

$$A = \{1, 2, 3, 4, 5\}.$$

Remark

- This method is suitable for sets with relatively few elements. When the set is large but follows a clear pattern, we use an ellipsis "..." to indicate continuation:

$$B = \{1, 2, 3, \ldots, 1000\}.$$

- If the pattern continues indefinitely, the set is **infinite**, for example:

$$\mathbb{N} = \{1, 2, 3, 4, 5, \ldots\},$$

  the meaning should be clear from the context.

- Order and repetition of elements do not matter. Then, the following sets describe the same set:

$$\{1, 2, 3\} = \{3, 2, 1\} = \{1, 2, 2, 3\}.$$

2. Set-Builder Form

In this representation, a set is described by a property that characterizes its elements rather than listing them explicitly:

$$A = \{x \mid P(x)\}.$$

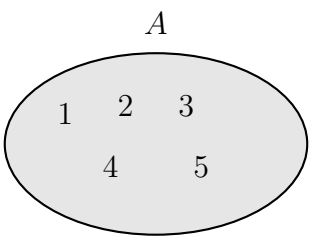
This reads: "A is the set of all elements x such that $P(x)$ is true." Set-builder form is also called implicit description.
We give the following example:

$$A = \{x \in \mathbb{R} \mid x \leq 5\}.$$

A is the set of real numbers that are less than or equal to 5.

3. Venn Diagram (Graphical Representation)

Sets can also be represented graphically using **Venn diagrams**. Each set is drawn as a closed curve (usually a circle or ellipse).
For example the set $A = \{1, 2, 3, 4, 5\}$ is represented with Venn Diagram as follows:



4. Interval Notation (for subsets of $\mathbb{R}$)

When dealing with subsets of $\mathbb{R}$, we often represent them using intervals:

$$[a, b] = \{x \in \mathbb{R} \mid a \leq x \leq b\},$$

$$]a, b[= \{x \in \mathbb{R} \mid a < x < b\},$$

$$[a, b[= \{x \in \mathbb{R} \mid a \leq x < b\},$$

$$]a, b] = \{x \in \mathbb{R} \mid a < x \leq b\}.$$

Square brackets "[ ]" indicate that the endpoint is **included**, while"] [" mean that the endpoint is **excluded**.

2.1.2 Types and Relations of Sets

Sets can be classified according to their elements and relationships with other sets.

Empty set

Definition

A set that does not contain any element is called an empty set. It is denoted by $\{\}$ or $\emptyset$.

Singleton set

Definition

If A is a set that contains a single element, then A is called a singleton and $|A| = 1$

Example
$A = \{2\}$ is a singleton.

Subset

Definition

We say that the set A is a subset of B denoted by $A \subseteq B$ if and only if every element of A is also an element of B. Formally this means: For any sets A and B,

$$A \subseteq B \quad \text{if and only if} \quad \forall x : x \in A \Rightarrow x \in B.$$

The relation $\subseteq$ is called the inclusion relation.

Remark

- The empty set is a subset of every set, including the empty set itself.
- The negation of $A \subseteq B$ is noted $A \nsubseteq B$,and we have

$$A \nsubseteq B \iff \overline{A \subseteq B}$$

  which is equivalent to

$$\exists x : x \in A \wedge x \notin B$$

- We note the proper inclusion by $\subset$ ($A \subset B$ i.e. all the elements of A are in B and $A \neq B$).

Example
Let $A = \{1, 2, 3\}$, $B = \{3, 2\}$ and $C = \{5\}$, we have $B \subset A$ and $C \nsubseteq A$

Equal Sets

Definition

Two sets A and B are said to be equal if they have exactly the same elements. Formally this means: For any sets A and B,

$$A = B \quad \text{if and only if} \quad \forall x : (x \in A \iff x \in B)$$

. Also we have

$$A = B \Leftrightarrow (A \subseteq B) \wedge (B \subseteq A)$$

Example
$A = \{1, 2, 3\}$ and $B = \{3, 2, 1\}$ are equal sets i.e. $(A \subseteq B) \wedge (B \subseteq A)$.

Equivalent Sets

Definition

Two sets A and B are said to be equivalent if they have the same number of elements i.e. $|A| = |B|$, even if the elements are different .

Example
$C = \{a, b, c\}$ and $A = \{1, 2, 3\}$ are equivalent sets because both have 3 elements, i.e. $|C| = |A| = 3$.

Power Set

Definition

Let A be a set, the power set of A is the set of all subsets of the set A. The power set of A is denoted by $\mathcal{P}(\mathbf{A})$. That is

$$\mathcal{P}(A) = \{X | X \subseteq A\}$$

We have

- $\emptyset \in \mathcal{P}(A)$ and $A \in \mathcal{P}(A)$
- $|\mathcal{P}(A)| = 2^n$ such as $|A| = n$

Example
Let the set $A = \{9, 1, 3\}$.

- We have $|A| = 3$ then $|\mathcal{P}(A)| = 2^3 = 8$.

and $\mathcal{P}(A) = \{\emptyset, \{9\}, \{1\}, \{3\}, \{9, 1\}, \{9, 3\}, \{1, 3\}, A\}$.

2.1.3 Set Operations

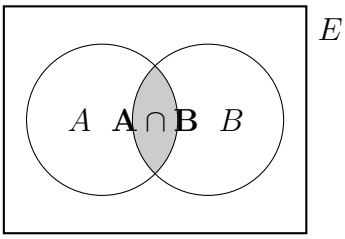
Let E be a set, and $A, B \in \mathcal{P}(E)$ throught all this part.

Intersection of Sets

Definition

The intersection of two sets A and B, denoted by $A \cap B$, is the set of all elements that belong to both A and B. Formally, we have

$$A \cap B = \{x \in E \mid x \in A \text{ and } x \in B\}.$$



Example
If $A = \{1, 2, 3\}$ and $B = \{3, 4, 5\}$, then $A \cap B = \{3\}$.

From the definition of the intersection we get

- $A \cap B \subseteq A$
- $A \cap B \subseteq B$

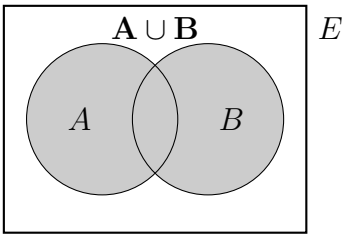
Union of Sets

Definition

The union of two sets A and B, denoted by $A \cup B$, is the set of all elements that belong to A or to B or to both. Formally, we have

$$A \cup B = \{x \in E \mid x \in A \text{ or } x \in B\}.$$

If A and B are finite sets, then we have $|A \cup B| = |A| + |B| - |A \cap B|$.



Example
If $A = \{1, 2, 3\}$ and $B = \{3, 4, 5\}$, then $A \cup B = \{1, 2, 3, 4, 5\}$.

From the definition of the union we get

- $A \subseteq A \cup B$
- $B \subseteq A \cup B$

Properties

1. **Commutative property**
   (a) $A \cup B = B \cup A$
   (b) $A \cap B = B \cap A$
2. **Associative property**
   (a) $A \cap (B \cap C) = (A \cap B) \cap C$
   (b) $A \cup (B \cup C) = (A \cup B) \cup C$
3. **Distributive property**
   (a) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
   (b) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

Complement of a Set

Definition

The complement of a set A relative to the universal set E denoted by $\complement_E A$ is the set of elements that are in E and not in A. That is,

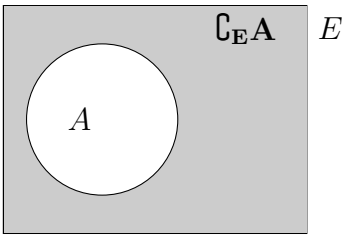
$$\complement_E A = \{x \in E \mid \ x \notin A\}$$

Furtheremore,

$$\forall x \in E : x \in \complement_E A \iff x \notin A$$

and

$$A \cup \complement_E A = E$$



If E is a finite set and A a subset of E, then:

$$A \cup \complement_E A = E \text{ and } |E| = |A| + |\complement_E A|$$

Example
If $E = \{1, 2, 3, 4, 5, 6\}$ and $A = \{1, 2, 3\}$, then $\complement_E A = \{4, 5, 6\}$ and $|\complement_E A| = |E| - |A| = 6 - 3 = 3$.

Properties

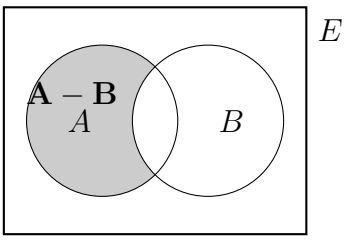
- $\complement_E(\complement_E A) = A$.
- $A \subseteq B \Leftrightarrow \complement_E B \subseteq \complement_E A$.
- $\complement_E(A \cup B) = \complement_E A \cap \complement_E B$. (De Morgan's Law)
- $\complement_E(A \cap B) = \complement_E A \cup \complement_E B$. (De Morgan's Law)

Difference of Sets

Definition

The difference of two sets A and B, denoted $A - B$, is the set of elements that belong to A and not to B.

$$A - B = \{x \in E \mid x \in A \text{ and } x \notin B\}.$$



Example
If $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6\}$, then $A - B = \{1, 2\}$.

Remark

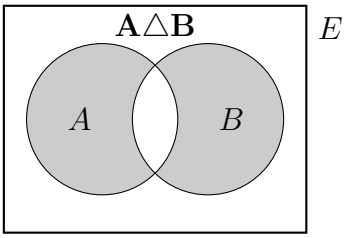
We have the complement of a set A relative to E is the difference of E and A, i.e. $\complement_E A = E - A$.

Symmetric Difference of Sets

Definition

The Symmetric difference of two sets A and B, denoted $A\Delta B$, is the set of elements that belong to A and not to B or that belong to B and not to A. Formally,

$$A\Delta B = (A - B) \cup (B - A)$$



Example
If $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6\}$, then the symmetric difference of A and B is

$$A\triangle B = (A - B) \cup (B - A) = \{1, 2, 5, 6\}.$$

Properties

1. $A\Delta B = (A \cup B) - (B \cap A)$
2. $A\Delta B = B\Delta A$

Cartesian product

Definition

The Cartesian product of two sets A and B denoted by $A \times B$ is the set of all ordered pairs whose first component belongs to A and second component belongs to B. That is

$$A \times B = \{(x, y) | x \in A \wedge y \in B\}.$$

If A and B are two finite sets, then the cardinality of their Cartesian product is given by:

$$|A \times B| = |A| \times |B|.$$

Example

If $A = \{1, 2\}$ and $B = \{7, 4, 1\}$, then

$$A \times B = \{(1, 7), (1, 4), (1, 1), (2, 7), (2, 4), (2, 1)\}.$$

and the cardinality is

$$|A \times B| = 2 \times 3 = 6.$$

2.2 Functions

2.2.1 Definitions and Operations

This part contains the definition of a function and other important definitions and the operations on functions.

Definition

A function f from a set E to a set F is a relation that associates each element of E (called the domain) with one and only one element of F (called the codomain), in other words, the assignment is unique in the sense that an $x \in E$ is assigned a unique element of F. Formally:

$$\forall x \in E, \exists! y \in F : y = f(x).$$

which is equuivalent to

$$\forall x, y \in E, x = y \Rightarrow f(x) = f(y),$$

we write:

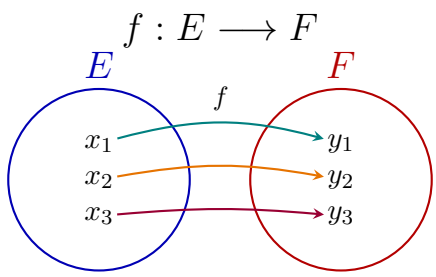
$$f : E \to F$$
$$x \mapsto y = f(x)$$

y is called the image of x under f and x is called the preimage of y.

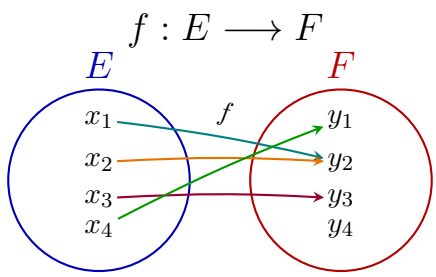
Remark

- Functions are also called Mappings or Maps.
- The set $\{y \in F : \text{There is an } x \in E \text{ such that } y = f(x)\}$ is called the range of f.
- The range of f is always a subset of the codomain.
- Note the distinction between $f(x)$ and f: $f(x)$ is a value in the codomain, which is the image of x, whereas f is the function that is the rule that assigns each preimage x its corresponding image.

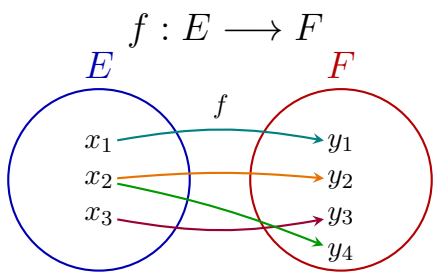
Representation of a function



Representation of a function



A function cannot assign two different images to the same element of E. The following representation is not a function.



Example

1. Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = x^2$. Here, each real number x has a unique image x^2.

2. Let $E = \{1, 2, 3, 4\}$ and $F = \{a, b, c\}$. A function $f : E \to F$ can be defined by specifying the image of each element of E:

$$f(1) = a, \quad f(2) = b, \quad f(3) = b, \quad f(4) = c.$$

   In this case, f is completely determined by these assignments, even though it is not given by an algebraic expression.

Definition

The graph of a function $f : E \to F$ noted $Graph(f)$ is a subset of the Cartesian product $E \times F$ defined by:

$$Graph(f) = \{(x, y) \in E \times F \mid y = f(x)\} = \{(x, f(x)) \mid x \in E\}$$

Definition

Let E be a non-empty set. The function from E to E that assigns each element $x \in E$ to itself is called the identity function on E, denoted by Id_E. It is defined as follows:

$$\mathrm{Id}_E : E \longrightarrow E$$

$$x \longmapsto x$$

Definition

Two functions $f : E \to F$ and $g : E' \to F'$ are said to be equal if and only if:

$$\begin{cases} E = E', \\ F = F', \\ \forall x \in E, \ f(x) = g(x). \end{cases}$$

Example

Let $f, g : \mathbb{R} \setminus \{-1\} \to \mathbb{R}$ be two functions defined by:

$$f(x) = \frac{x^2 - 1}{x + 1} \quad \text{and} \quad g(x) = x - 1$$

For all $x \in \mathbb{R} \setminus \{-1\}$, we have:

$$f(x) = \frac{(x - 1)(x + 1)}{x + 1} = x - 1 = g(x)$$

Also f and g have the same domain and codomain.
Therefore, $f = g$.

Operations on Functions

Addition: Let $f, g : E \to F$ be two functions. Their sum is the function $f + g : E \to F$ defined by:

$$(f + g)(x) = f(x) + g(x)$$

Product: Let $f, g : E \to F$ be two functions. Their product is the function $f \times g : E \to F$ defined by:

$$(f \times g)(x) = f(x) \times g(x)$$

Composition: Let $f : E \to F$ and $g : F \to G$. The composition of f and g, denoted $g \circ f$, is the function from E to G defined by:

$$(g \circ f)(x) = g(f(x)), \quad \forall x \in E.$$

It satisfies:

$$Id_F \circ f = f \circ Id_E = f$$

$$h \circ (g \circ f) = (h \circ g) \circ f \quad \text{(associativity)}$$

In general, composition is **not commutative**:

$$f \circ g \neq g \circ f$$

2.2.2 Direct Image and Inverse Image

Direct Image

Definition

Let $f : E \to F$ be a function and $A \subseteq E$. The direct image of A by f, denoted by $f(A)$, is the subset of F containing the images of the elements of A by f.

$$f(A) = \{y \in F \mid \exists x \in A : f(x) = y\} = \{f(x) \mid x \in A\} \subseteq F$$

Furtheremore,

$$\forall y \in F : \; y \in f(A) \iff \exists x \in A : f(x) = y.$$

Example

We consider the function $f : \mathbb{R} \to \mathbb{R}$ defined by:

$$\forall x \in \mathbb{R},\ f(x) = 2x - 1.$$

Find $f([0,1])$.

$$f([0,1]) = \{f(x) \in \mathbb{R} \mid x \in [0,1]\}.$$

Let $y \in f([0,1])$, then

$$y \in f([0,1]) \Leftrightarrow \exists x \in [0,1] :\ y = f(x) = 2x - 1.$$

We have $x \in [0,1]$. We note that f is monotonic (strictly increasing) on $\mathbb{R}$, hence on $[0,1]$. Thus, for $x \in [0,1]$ we obtain

$$0 \leq x \leq 1 \Rightarrow -1 \leq 2x - 1 \leq 1.$$

Therefore,

$$y = 2x - 1 \in [-1,1].$$

Hence $f([0,1]) \subseteq [-1,1]$.

Conversely, we prove $\forall y \in [-1,1], \exists x \in [0,1] :\ y = f(x) = 2x - 1$.
For any $y \in [-1,1]$, we solve $y = 2x - 1$, i.e. $x = \dfrac{y+1}{2}$. Since $-1 \leq y \leq 1$, we have

$$0 \leq \frac{y+1}{2} \leq 1,$$

hence

$$x = \frac{y+1}{2} \in [0,1].$$

Therefore, for every $y \in [-1,1]$ we have $x = \dfrac{y+1}{2} \in [0,1]$ such that $f\left(\dfrac{y+1}{2}\right) = y$.

Hence $[-1,1] \subseteq f([0,1])$.
Combining the two inclusions gives $f([0,1]) = [-1,1]$.

Inverse Image (Preimage)

Definition

Let $f : E \to F$ be a function and $B \subseteq F$. The inverse image of B by f, denoted by $f^{-1}(B)$, is the subset of E containing the preimages of the elements of B by f.

$$f^{-1}(B) = \{x \in E \mid f(x) \in B\} \subseteq E$$

Also, we have

$$\forall x \in E :\ x \in f^{-1}(B) \iff f(x) \in B.$$

Example

We consider the function $f : \mathbb{R} \to \mathbb{R}$ defined by:

$$\forall x \in \mathbb{R},\ f(x) = x^2.$$

For $B = \left\{\frac{1}{16}\right\} \subset \mathbb{R}$, what is $f^{-1}(B)$?

Solution:

By definition, the inverse image is:

$$f^{-1}(B) = \{x \in \mathbb{R} \mid f(x) \in B\} = \left\{x \in \mathbb{R} \mid x^2 \in \left\{\frac{1}{16}\right\}\right\}$$

We find the values of x by solving the following equation:

$$x^2 = \frac{1}{16}.$$

This gives:

$$x = \pm\frac{1}{4}.$$

Thus, the inverse image is:

$$f^{-1}(B) = \left\{-\frac{1}{4}, \frac{1}{4}\right\}.$$

Properties of Direct and Reciprocal Images

Let $f : E \longrightarrow F$ be a function. For $A, B \subseteq E$ and $C, D \subseteq F$ we have:

1. $A \subseteq B \Rightarrow f(A) \subseteq f(B)$.
2. $f(A \cup B) = f(A) \cup f(B)$.
3. $f(A \cap B) \subseteq f(A) \cap f(B)$.
4. $C \subseteq D \Rightarrow f^{-1}(C) \subseteq f^{-1}(D)$.
5. $f^{-1}(C \cup D) = f^{-1}(C) \cup f^{-1}(D)$.
6. $f^{-1}(C \cap D) = f^{-1}(C) \cap f^{-1}(D)$.

2.2.3 Injection, Surjection, Bijection

Definition

A function $f : E \to F$ is said to be injective (or one-to-one) if every distinct elements of the domain have distinct images. i.e. if every element of F has at most one preimage (or none) by f. Formally

$$\forall x_1, x_2 \in E : \; f(x_1) = f(x_2) \Rightarrow x_1 = x_2.$$

Equivalently,

$$\forall x_1, x_2 \in E : \; x_1 \neq x_2 \Rightarrow f(x_1) \neq f(x_2).$$

Example

$f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = 2x + 3$ is injective on $\mathbb{R}$ because we have for all x_1, $x_2 \in \mathbb{R}$, $f(x_1) = f(x_2) \Rightarrow 2x_1 + 3 = 2x_2 + 3 \Rightarrow 2x_1 = 2x_2 \Rightarrow x_1 = x_2$ then f is injective.

Definition

A function $f : E \to F$ is said to be surjective (or onto) if every element of F is the image of at least one element of E, that is:

$$\forall y \in F, \; \exists x \in E \text{ such that } f(x) = y.$$

Example

Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = 2x - 3$. For every $y \in \mathbb{R}$, we can find $x = \dfrac{y+3}{2} \in \mathbb{R}$ such that $f(x) = y$. Therefore, f is a surjective function since each real number y is the image of some real number x.

Definition

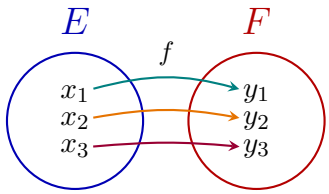
A function $f : E \to F$ is said to be bijective if it is both injective and surjective, i.e., if it establishes f is onto and one-to-one. Formally:

$$\forall y \in F, \; \exists!\, x \in E \text{ such that } f(x) = y.$$

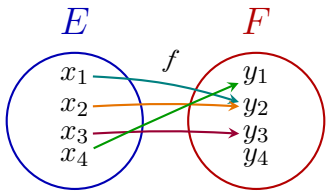
Example

$f(x) = x + 1$ from $\mathbb{R}$ to $\mathbb{R}$ is surjective and injective then it is bijective.

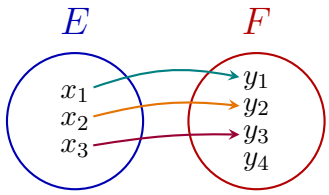
We have the following representations to better understand the previous properties:



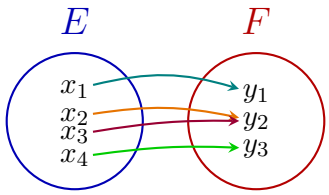
Bijective i.e. Injective and Surjective



Not Injective and Not Surjective



Injective and Not Surjective



Surjective and Not Injective

Remark

The negations of the properties of functions can be expressed as follows:

- f is not injective $\iff \exists x_1, x_2 \in E,\ x_1 \neq x_2$ and $f(x_1) = f(x_2)$.
- f is not surjective $\iff \exists y \in F, \forall x \in E:\ f(x) \neq y$.
- f is not bijective $\iff$ f is not injective or not surjective.

2.2.4 Inverse Function

Definition

If $f : E \to F$ is bijective, then we can define a function $g : F \to E$ such as for all $y \in F$; $g(y)$ is the unique preimage of y by g. The function g noted f^{-1} and we write:

$$f^{-1} : F \to E$$

$$y \mapsto x = f^{-1}(y)$$

such that:

$$f^{-1}(f(x)) = x, \quad \forall x \in E, \quad \text{and} \quad f(f^{-1}(y)) = y, \quad \forall y \in F.$$

The function f^{-1} is called the inverse function of f.

Example

Let $f : \mathbb{R} \to \mathbb{R}$ be defined by

$$f(x) = 2x + 3.$$

We start by proving that f is bijective.
(a) Injectivity: Let $x_1, x_2 \in \mathbb{R}$ such that $f(x_1) = f(x_2)$. Then:

$$2x_1 + 3 = 2x_2 + 3.$$

Subtracting 3 from both sides gives:

$$2x_1 = 2x_2 \quad \Rightarrow \quad x_1 = x_2.$$

Hence, f is **injective**.

(b) Surjectivity: Let $y \in \mathbb{R}$. We need to find $x \in \mathbb{R}$ such that $f(x) = y$. We have:

$$y = 2x + 3 \quad \Rightarrow \quad x = \frac{y-3}{2}.$$

Since $\frac{y-3}{2} \in \mathbb{R}$ for every $y \in \mathbb{R}$, such an x always exists. Thus, f is **surjective**.
Since f is both injective and surjective, it is therefore **bijective** hence the inverse function of f exists.
To find f^{-1}, we start from:

$$y = 2x + 3.$$

We already solved for x in terms of y:

$$x = \frac{y-3}{2} \in \mathbb{R}.$$

because f is bijective then x is unique.
We give the inverse function:

$$f^{-1} : F \to E$$
$$y \mapsto x = f^{-1}(y) = \frac{y-3}{2}$$

Verification:

$$f(f^{-1}(y)) = 2\left(\frac{y-3}{2}\right) + 3 = y, \quad \text{and} \quad f^{-1}(f(x)) = \frac{(2x+3)-3}{2} = x.$$

Hence, f^{-1} is indeed the inverse of f.

2.2.5 Restriction and Extension of a Function

Definition

Let $f : E \to F$ be a function and $A \subseteq E$. Then the function $g : A \to F$ defined by

$$g(x) = f(x) \quad \forall x \in A$$

is said to be the restriction function of f and is denoted by

$$g = f|_A.$$

Example

Let $f : \mathbb{R} \to \mathbb{R}$ be defined by

$$f(x) = x^2.$$

We consider the subset $A = [0, +\infty[\subseteq \mathbb{R}$. The function

$$g : A \to \mathbb{R}, \quad g(x) = f(x) = x^2$$

is the restriction of f to A, denoted $g = f|_A$.
Thus g has domain $A = [0, +\infty[$, while f is defined on $\mathbb{R}$.

Definition

Let $f : A \to F$ be a function and $A \subseteq E$. Then the function $g : E \to F$ defined by

$$g(x) = f(x) \quad \forall x \in A$$

is said to be the extension function of f from A to E.

Example

Let $f : [0, +\infty[\to \mathbb{R}$ be defined by

$$f(x) = \sqrt{x}.$$

We can define an extension $g : \mathbb{R} \to \mathbb{R}$ of f by

$$g(x) = \begin{cases} \sqrt{x}, & x \geq 0, \\ 0, & x < 0. \end{cases}$$

Then g coincides with f on $[0, +\infty[$, so g is an extension of f from $[0, +\infty[$ to $\mathbb{R}$.

Exercise 9: Sets Operations

Let $A = \{\{-2\}, \{3\}, \{4\}\}$ and $B = \{\{3\}, \{-2, 3, 4\}\}$.

1. Determine the following sets:
   $A \cap B, \ A \cup B, \ A \Delta B, \ B - A, \ \mathcal{P}(A), \ A \times B.$
2. Replace the dots by one of the following symbols: $\in$, $\subset$, or $\notin$.
   $\{3\} \ldots A \cap B$, $\{\{-2\}\} \ldots A \cup B$, $\{-2, 3, 4\} \ldots A \Delta B$, $\{-2\} \ldots B - A$, $\{A\} \ldots \mathcal{P}(A)$, $(\{-2\}, \{5\}) \ldots A \times B$.

3. Let the universal set be $E = A \cup (B \cup \{\{5\}\})$.
   (a) Determine $\complement_E A$.
   (b) Let E be a set and $A_1, A_2, \ldots, A_n$ be subsets of E. We say that these subsets form a partition for the set E (i.e. $\{A_1, A_2, \ldots, A_n\}$ is a partition for E) if the following three conditions are satisfied:
      i. If $i, j \in \{1, 2, \ldots, n\}$ and $i \neq j$, then $A_i \cap A_j = \varnothing$.
      ii. For all $i \in \{1, 2, \ldots, n\}$, $A_i \neq \varnothing$.
      iii. $E = A_1 \cup A_2 \cup \cdots \cup A_n$.
      - Using the previous definition find a partition for the set $E = A \cup (B \cup \{\{5\}\})$.

Test Your Chapter Mastery

Exercise 10: Sets Representations

1. We consider the set

$$E = \{x \in \mathbb{Z} \mid x^2 - 5x + 6 = 0 \text{ or } -2 < x \leq 2\}.$$

   - Write E in the roster form (Listing explicitly the elements of E).
   - We consider another finite set F such that:

$$|\mathcal{P}(F)| = 32 \quad \text{and} \quad |E \cup F| = 9.$$

   (a) Find $|E \cap F|$.

2. Let the following sets be given:

$$A = \{(x, y) \in \mathbb{R}^2 \mid 4x - y = 1\} \quad \text{and} \quad C = \{(t + 1, 4t + 3) \mid t \in \mathbb{R}\}.$$

   - Prove that $A = C$.

3. Let

$$D = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 1\}.$$

   - Show that D cannot be written as the Cartesian product of two subsets of $\mathbb{R}$.

Exercise 11: Sets Properties

1. Let A, B, and C are subsets of a universal set E
   (a) Simplify the following sets:

$$(\complement_E(A \cup B)) \cap (\complement_E(C \cup \complement_E A)), \qquad (\complement_E(A \cap \complement_E B)) \cup (\complement_E(C \cap \complement_E A)).$$

(b) Prove the following properties:

- $A \subseteq B \Leftrightarrow \complement_E B \subseteq \complement_E A$.
- $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.
- If $A \cap B = A \cup B$, then $A = B$.
- $A - B = A \cap \complement_E B$.
- $A \cap (B - C) = (A \cap B) - (A \cap C)$.
- $A \cap (B \triangle C) = (A \cap B) \triangle (A \cap C)$.

(c) Put $\subseteq$ or $=$ between the following sets(Justify):

i. $\mathcal{P}(A \cup B)$ and $\mathcal{P}(A) \cup \mathcal{P}(B)$?

ii. $\mathcal{P}(A \cap B)$ and $\mathcal{P}(A) \cap \mathcal{P}(B)$?

You can use the property $(A \subseteq B) \implies (\mathcal{P}(A) \subseteq \mathcal{P}(B))$

2. E denotes the universal finite set and let $A, B, C, D \subseteq E$. Prove that

$$|\complement_E\Big((A \times B) \triangle (C \times D)\Big)| = |E|^2 - |A| \times |B| - |C| \times |D| + 2|(A \times B) \cap (C \times D)|.$$

You may use the following properties:

$$|A \cup B| = |A| + |B| - |A \cap B|, \quad |\complement A| = |E| - |A|, \quad |A - B| = |A| - |A \cap B|, \quad |A \times B| = |A| \times |B|$$

Exercise 12: Direct and Inverse Images

1. Let $f : E \longrightarrow F$ be a function. For $A, B \subseteq E$ and $C, D \subseteq F$. Prove the following properties:

- $f(A \cup B) = f(A) \cup f(B)$.
- If f is injective then $f(A \cap B) = f(A) \cap f(B)$.
- $f^{-1}(C \cup D) = f^{-1}(C) \cup f^{-1}(D)$.
- $f^{-1}(C \cap D) = f^{-1}(C) \cap f^{-1}(D)$.

2. Let

$$f : \mathbb{R} \longrightarrow \mathbb{R}, \qquad f(x) = x^2 + 1.$$

(a) Determine $f([-2, 1])$, $f([-2, 1] \cup [2, +\infty[)$ and $f(\mathbb{R})$.

(b) Determine $f^{-1}([0, 1])$, $f^{-1}(]-\infty, 5] \cup [10, +\infty[)$ and $f^{-1}([0, 1] \cap [-4, 0[)$.

(c) f is injective? Give a restriction of f that is injective (Justify your answer).

(d) We define another function

$$g : \mathbb{R} \longrightarrow \mathbb{R}, \qquad g(x) = 3x - 1.$$

Determine the following functions (compositions):

$$g \circ f \quad \text{and} \quad f \circ g,$$

with their explicit expressions. Are these two compositions equal?

Exercise 13: Function on Finite Sets

Let

$$E = \{m, n, p, q, r, s\}, \qquad F = \{1, 2, 3, 4, 5, 6\}.$$

We define a function $f : E \to F$ as follows:

$$f(m) = 2, \qquad f(p) = 5, \qquad f(r) = 1.$$

For the other elements of E, we know that

$$\{(n, 4), (q, 2), (s, 6)\} \subset Graph(f).$$

1. Determine the direct image of the subset $A = \{m, n, q, s\}$.
2. Determine the inverse image of the subset $B = \{2, 3, 4, 6\}$.
3. Determine the range of f.
4. We have the following property that is true for any function f defined from a set E to a set F

   $$f \text{ is surjective} \Leftrightarrow f(E) = F.$$

   Using this property answer the following question: Is the function f –defined in the exercise– surjective?

Exercise 14: Inverse function

Let f be a function defined by:

$$f : \mathbb{R} - \left\{\frac{1}{2}\right\} \longrightarrow \mathbb{R},$$

$$x \longmapsto y = f(x) = \frac{x+1}{2x-1}.$$

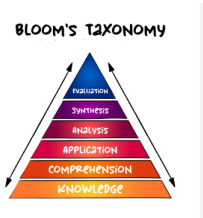
1. Prove that f is a well-defined function.
2. Show that f is injective. Is f bijective?
3. What restriction should we impose on the codomain so that f becomes a bijection?
4. Find the inverse function f^{-1} of f.

3

Binary Relations on a set

Specific Objectives

By the end of this chapter, students will be able to:



1. Define the key concepts of binary relations, including reflexive, symmetric, transitive, and antisymmetric properties.
2. Explain clearly the differences between equivalence relations and order relations (partial and total).
3. Apply the properties of binary relations, including the determination of equivalence classes, to solve classification and verification problems.
4. Analyze a binary relation represented as a graph to determine its structural properties.
5. Evaluate whether a given binary relation satisfies the conditions of an equivalence relation or an order relation.
6. Construct a binary relation on a set with its specific properties.

Introduction

A binary relation on a set describes how elements may be connected or compared. By analyzing properties such as reflexivity, symmetry, and transitivity, we classify relations into equivalence and order relations.
These structures are fundamental in computer science, where they support database modeling, sorting algorithms, and state-transition systems.
This chapter is divided into three sections: the first focuses on the definition of a binary relation on a set and its properties, the second on order relation and the third on equivalence relation.

3.1 Basic Definitions

3.1.1 Definition of a Binary Relation

Definition

Let E be a non-empty set. A binary relation on the set E noted $\mathcal{R}$ is a subset of the Cartesian product $E \times E$; that is,

$$\mathcal{R} \subseteq E \times E.$$

If $\mathcal{R}$ is a binary relation on E, then for all $x, y \in E$,

$$(x, y) \in \mathcal{R} \quad \Longleftrightarrow \quad x \,\mathcal{R}\, y.$$

In this case, we say that x *is related to* y by the binary relation $\mathcal{R}$ on the set E.

Remark

- If $(x, y) \notin R$, we write $x \not\mathcal{R} y$ and say that x is not related to y by the relation R.
- If $\mathcal{R}$ is a binary relation on E. For all $(x, y) \in E \times E$, we have:

$$x\mathcal{R}y \quad \text{or} \quad x \not\mathcal{R} y.$$

  This means that every pair (x, y) either belongs to the relation $\mathcal{R}$ or does not.

A binary relation $\mathcal{R}$ on a set E can be represented graphically as a directed graph in which:

- each element of E is represented by a vertex,
- and there is a directed edge from x to y whenever $(x, y) \in \mathcal{R}$.

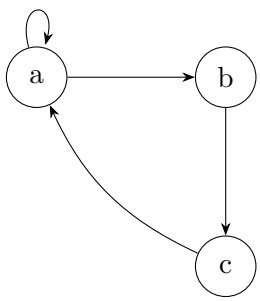


Figure 3.1 – Representation of a binary relation $R = \{(a, a), (a, b), (b, c), (c, a)\}$ on the set $E = \{a, b, c\}$.

Example

Let $E = \{1, 2, 3\}$, and we define

$$\mathcal{R} = \{(1, 1), (1, 2), (2, 3)\}.$$

Then $\mathcal{R}$ is a binary relation on E; We have $1\mathcal{R}2$ holds, but $2 \not\mathcal{R} 1$.

3.1.2 Properties of Binary Relations

Reflexive Relation

Definition

A binary relation $\mathcal{R}$ on a set E is called reflexive if it satisfies the following property.

$$\forall x \in E : \ x\mathcal{R}x.$$

Symmetric Relation

Definition

We say that a binary relation $\mathcal{R}$ on a set E is symmetric if it satisfies the following property.

$$\forall x, y \in E : \ (x\mathcal{R}y) \Rightarrow (y\mathcal{R}x).$$

Antisymmetric Relation

Definition

A binary relation $\mathcal{R}$ on a set E is called antisymmetric if it satisfies the following property.

$$\forall x, y \in E : \ (x\mathcal{R}y) \wedge (y\mathcal{R}x) \Rightarrow x = y.$$

Transitive Relation

Definition

We say that a binary relation $\mathcal{R}$ on a set E is transitive if it satisfies the following property.

$$\forall x, y, z \in E : \ (x\mathcal{R}y) \wedge (y\mathcal{R}z) \Rightarrow (x\mathcal{R}z).$$

Remark

- To prove that $\mathcal{R}$ is not reflexive, we show the following property.

$$\exists x \in E : \ x \not\mathcal{R} x.$$

- To show that $\mathcal{R}$ is not symmetric, we prove that there exist $x, y \in E$ such that

$$x\mathcal{R}y \text{ and } y \not\mathcal{R} x.$$

- To prove that $\mathcal{R}$ is not antisymmetric, we prove the following property.

$$\exists x, y \in E : \ x \neq y \text{ and } x\mathcal{R}y \text{ and } y\mathcal{R}x.$$

- To prove that $\mathcal{R}$ is not transitive, we show that the following property is true.

$$\exists x, y, z \in E : \ x\mathcal{R}y \text{ and } \ y\mathcal{R}z \text{ and } x \not\mathcal{R} z.$$

Example

Let $\mathcal{R}$ be the binary relation on $\mathbb{R}$ defined by

$$x\mathcal{R}y \iff xe^y = ye^x.$$

We verify the four properties of a binary relation.

Reflexive: We verify if the following property "$\forall x \in \mathbb{R} : \ x\mathcal{R}x$" is true or false. Let $x \in \mathbb{R}$, we have $xe^x = xe^x$, hence $x\mathcal{R}x$. Therefore, $\mathcal{R}$ is reflexive.

Symmetric: If it is symmetric we must prove the following property "$\forall x, y \in \mathbb{R} : (x\mathcal{R}y) \Rightarrow (y\mathcal{R}x)$"is true, if it is false we must find $x, y \in \mathbb{R}$ such that $x\mathcal{R}y$ and $y\not\mathcal{R}x$. Let $x, y \in \mathbb{R}$. we suppose that $x\mathcal{R}y$, then $xe^y = ye^x$, which also gives $ye^x = xe^y$, so $y\mathcal{R}x$. Therefore, $\mathcal{R}$ is symmetric.

Transitive: Let $x, y, z \in \mathbb{R}$ such that $x\mathcal{R}y$ and $y\mathcal{R}z$. Then $xe^y = ye^x$ and $ye^z = ze^y$. Compute xye^z:

$$xye^z = x(ye^z) = x(ze^y) = z(xe^y) = z(ye^x) = yze^x.$$

Hence $xye^z = yze^x$. If $y \neq 0$, dividing by y gives $xe^z = ze^x$, and thus $x\mathcal{R}z$. For the case $y = 0$, we have $x = 0$ and $z = 0$, so $x\mathcal{R}z$ as well. Therefore, $\mathcal{R}$ is transitive.

Antisymmetric: The relation is not antisymmetric, because "$\forall x, y \in \mathbb{R} : (x\mathcal{R}y) \wedge (y\mathcal{R}x) \Rightarrow x = y$" is false. We prove that is false by finding a counter example. Let

$$x = \frac{e}{e-1}, \qquad y = \frac{1}{e-1}.$$

Then

$$xe^y = \frac{e}{e-1}\, e^{\frac{1}{e-1}} \quad \text{and} \quad ye^x = \frac{1}{e-1}\, e^{\frac{e}{e-1}}.$$

Since $e^{\frac{e}{e-1}} = e^{\frac{1}{e-1}}\, e$, both sides are equal: $xe^y = ye^x$. Hence $x\mathcal{R}y$ and $y\mathcal{R}x$. However, $x \neq y$, so $\mathcal{R}$ is not antisymmetric.

Therefore, the relation $\mathcal{R}$ is reflexive, symmetric, and transitive, but not antisymmetric.

3.2 Order Relation

3.2.1 Definition

Definition

A binary relation $\mathcal{R}$ on a set E is called an order relation if $\mathcal{R}$ is reflexive, transitive and antisymmetric. The pair $(E, \mathcal{R})$ is called an ordered set.

3.2.2 Partial and Total Order

Definition

An order relation $\mathcal{R}$ on a set E is a total order if for all $x, y \in E$, we have

$$x\mathcal{R}y \text{ or } y\mathcal{R}x.$$

Definition

Let $\mathcal{R}$ be a total order relation on a set E. The pair $(E, \mathcal{R})$ is called a totally ordered set.

Definition

An order relation $\mathcal{R}$ on a set E is a partial order if there exist $x, y \in E$, such that

$$x \not\mathcal{R} y \wedge y \not\mathcal{R} x.$$

Definition

Let $\mathcal{R}$ be a partial order relation on a set E. The pair $(E, \mathcal{R})$ is called a partially ordered set.

Example

A relation $\ll$ on $\mathbb{N}^*$ is defined for all $x, y \in \mathbb{N}^*$:

$$x \ll y \iff \exists n \in \mathbb{N}^* \text{ such that } y = x^n$$

Proving $\ll$ is an Order Relation:

For $\ll$ to be an order relation, we must prove the three properties: Reflexivity, Antisymmetry, and Transitivity.

1. Reflexivity: For every $x \in \mathbb{N}^*$, we must have $x \ll x$.
Let $x \in \mathbb{N}^*$, we need to find an exponent $n \in \mathbb{N}^*$ such that $x = x^n$. This condition is satisfied by choosing $n = 1$. Since $1 \in \mathbb{N}^*$, we have $x = x^1$, which means $x \ll x$. Thus, the relation $\ll$ is reflexive.

2. Antisymmetry: For every $x, y \in \mathbb{N}^*$, if $x \ll y$ and $y \ll x$, then $x = y$.
Let $x, y \in \mathbb{N}^*$, we assume that $x \ll y$ and $y \ll x$. By definition:

1. $x \ll y \iff \exists n \in \mathbb{N}^*$ such that $y = x^n$.
2. $y \ll x \iff \exists m \in \mathbb{N}^*$ such that $x = y^m$.

Substituting (1) into (2) gives: $x = (x^n)^m = x^{nm}$. Since $x \in \mathbb{N}^*$, we analyze the two possibilities for x:

- If $\mathbf{x = 1}$, then from (1), $y = 1^n = 1$. Hence, $x = y$.
- If $\mathbf{x > 1}$, then the equality $x^1 = x^{nm}$ implies that the exponents must be equal: $1 = nm$. Since $n, m \in \mathbb{N}^*$, the only solution is $n = 1$ and $m = 1$. Substituting $n = 1$ back into (1) yields $y = x^1 = x$. Hence, $x = y$.

In all cases, $x = y$. Thus, the relation $\ll$ is antisymmetric.

3. Transitivity: For every $x, y, z \in \mathbb{N}^*$, if $x \ll y$ and $y \ll z$, then $x \ll z$.
Let $x, y, z \in \mathbb{N}^*$, we assume that $x \ll y$ and $y \ll z$. By definition:

1. $x \ll y \iff \exists n \in \mathbb{N}^*$ such that $y = x^n$.
2. $y \ll z \iff \exists m \in \mathbb{N}^*$ such that $z = y^m$.

Substituting the expression for y from (1) into (2):

$$z = (x^n)^m = x^{nm}$$

Let $k = nm$. Since $n, m \in \mathbb{N}^*$, their product $k \in \mathbb{N}^*$. Thus, $z = x^k$ with $k \in \mathbb{N}^*$, which means $x \ll z$. Thus, the relation $\ll$ is transitive.
Since the relation $\ll$ satisfies Reflexivity, Antisymmetry, and Transitivity, it is an order relation.
Proving the Order is not total
We must find $x, y \in \mathbb{N}^*$ such that $x \not\ll y$ and $y \not\ll x$.
We consider the elements $x = 2$ and $y = 3$.

- **Test $2 \ll 3$:** This requires $3 = 2^n$ for some $n \in \mathbb{N}^*$. Since $2^1 = 2$ and $2^2 = 4$, there is no $n \in \mathbb{N}^*$ such that $2^n = 3$. Therefore, $2 \not\ll 3$.
- **Test $3 \ll 2$:** This requires $2 = 3^n$ for some $n \in \mathbb{N}^*$. Since $3^1 = 3$, and for all $n > 1$, $3^n > 2$, there is no $n \in \mathbb{N}^*$ such that $3^n = 2$. Therefore, $3 \not\ll 2$.

Since we have found elements 2 and 3 that are non-comparable, the order relation $\ll$ is **not total**.
Therefore, the relation $\ll$ is a partial order relation on $\mathbb{N}^*$.

3.3 Equivalence Relation

3.3.1 Definition of an Equivalence Relation

Definition

A binary relation $\mathcal{R}$ on a set E is said to be an equivalence relation if $\mathcal{R}$ is reflexive, symmetric and transitive.

Example

We define the binary relation R on $\mathbb{C}$ by:

$$\forall z, z' \in \mathbb{C}, \quad z\mathcal{R}z' \Leftrightarrow |z| = |z'|.$$

Question: Is R an equivalence relation?

- Reflexive: Let $z \in \mathbb{C}$, we have $|z| = |z|$ hence $z\mathcal{R}z$, so R is reflexive.
- Symmetric: We have for all $z, z' \in \mathbb{C}$

  $$z\mathcal{R}z' \Leftrightarrow |z| = |z'| \Rightarrow |z'| = |z| \Rightarrow z'\mathcal{R}z,$$

  so R is symmetric.
- Transitive: Let $z, z', z'' \in \mathbb{C}$, we suppose that $z\mathcal{R}z'$ and $z'\mathcal{R}z''$ we get

  $$|z| = |z'| \ \wedge |z'| = |z''|,$$

  thus,

  $$|z| = |z''|$$

  hence

  $$(z\mathcal{R}z' \wedge z'\mathcal{R}z'') \Rightarrow z\mathcal{R}z'',$$

  so $\mathcal{R}$ is transitive.

Therefore, $\mathcal{R}$ is reflexive, symmetric and transitive, so $\mathcal{R}$ is an equivalence relation.

3.3.2 Equivalence Class

Definition

Let R be an equivalence relation on a set E. For every element $x \in E$ we define the equivalence class of x denoted by $\overline{x}$ as the set of all the elements of the set E that are related to x under the relation $\mathcal{R}$, i.e.

$$\overline{x} = \{y \in E \mid x\mathcal{R}y\}.$$

Definition

Let R be an equivalence relation on a set E. A quotient set noted $E/\mathcal{R}$ is the set of all equivalence classes of the set E. Formally

$$E/\mathcal{R} = \{\overline{x} \mid x \in E\}$$

Example
We already proved that the binary relation $\mathcal{R}$ defined on $\mathbb{C}$ by:

$$\forall z, z' \in \mathbb{C}, \quad z\mathcal{R}z' \Leftrightarrow |z| = |z'|.$$

is an equivalence relation.
Determine the equivalence class of every $z \in \mathbb{C}$ and the quotient set $\mathbb{C}/\mathcal{R}$.

Equivalence class: For the relation $z\mathcal{R}z' \iff |z| = |z'|$, the equivalence class of z is denoted by $\overline{z}$:

$$\overline{z} = \{\, w \in \mathbb{C} : |w| = |z| \,\}.$$

Every nonzero complex number can be written in exponential form $w = |z|e^{i\theta}$, $\theta \in \mathbb{R}$. Hence:

$$\overline{z} = \{\, |z|e^{i\theta} : \theta \in \mathbb{R} \,\}.$$

In particular, $\overline{0} = \{0\}$. Although the complex number 0 cannot be written in the usual polar form $re^{i\theta}$ because its argument is not defined, the expression $|0|e^{i\theta}$ always equals 0. Therefore the equivalence class of 0 is still $\{0\}$, and we may use the general formula

$$\overline{z} = \{\, |z|e^{i\theta} : \theta \in \mathbb{R} \,\}$$

for all $z \in \mathbb{C}$, including $z = 0$.

Quotient set: The quotient set is the collection of all such classes:

$$\mathbb{C}/\mathcal{R} = \{\, \overline{z} \mid \ z \in \mathbb{C} \,\} = \Big\{ \{re^{i\theta} : \theta \in \mathbb{R}\} \mid r \in [0, \infty) \Big\}.$$

Test Your Chapter Mastery

Exercise 15: Order Relation

We define on $\mathbb{R}^2$ the binary relation denoted by $\prec$ defined as follows:

$$\forall (x, y) \in \mathbb{R}^2 : \ (x, y) \prec (x', y') \iff x \leq x' \text{ and } y \leq y'.$$

1. Prove that $\prec$ is an order relation on $\mathbb{R}^2$.
2. Is the order total or partial?

Exercise 16: Equivalence Relation

Let E be a set and let $\mathcal{P}(E)$ denote the power set of E. We define a binary relation $\mathcal{R}$ on $\mathcal{P}(E)$ by

$$A\mathcal{R}B \quad \iff \quad A = B \text{ or } A = \complement_E B,$$

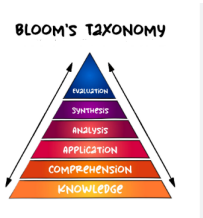
1. Prove that $\mathcal{R}$ is an equivalence relation on $\mathcal{P}(E)$.
2. Determine the equivalence class of an arbitrary subset $A \subseteq E$.

4

Algebraic Structures

Specific Objectives

By the end of this chapter, students will be able to:



1. Recall the definition of an internal composition law.
2. Explain the difference between a group and an abelian group.
3. Apply group axioms to verify whether a given set equipped with an internal composition law form a group.
4. Analyze the relation between the different structures groups, rings and fields.
5. Evaluate whether a function between algebraic structures is a homomorphism.
6. Create simple examples of finite rings or fields.

Introduction

Algebraic structures provide a formal framework for studying sets equipped with operations that satisfy specific properties. They allow us to understand groups, rings, and fields, which model symmetry, computation, and data manipulation. In computer science, these structures are essential in cryptography, automata theory, coding theory, and the design of efficient algorithms. By mastering algebraic structures, students develop tools that connect abstract mathematics to practical computational problems. This chapter is divided into four sections: the first focuses on internal composition law and its properties, the second on groups structure, the third on rings and finally the fourth focuses on the structure of fields.

4.1 Internal Composition Law

4.1.1 Definition and Examples

Definition

Let E be a non-empty set. An internal composition law (binary operation) on E is a function from $E \times E$ to E, if we denoted by $*$, this function assigns to every ordered pair $(x, y) \in E \times E$ an element $x * y$ that still belongs to E, we say that $x * y$ is the composition of x and y for the law $*$.

$$E \times E \longrightarrow E$$
$$(x, y) \mapsto x * y$$

Remark

The symbol $*$ is merely a notation for the internal composition law. In fact, any symbol such as $\top$, $\perp$, $\oplus$, or $\odot$ could be used to denote this operation.

Examples

1. On $E = \mathbb{Z}$, the addition, defined by

$$\mathbb{Z} \times \mathbb{Z} \longrightarrow \mathbb{Z} \quad (a, b) \mapsto a + b$$

the multiplication, defined by

$$\mathbb{Z} \times \mathbb{Z} \longrightarrow \mathbb{Z} \quad (a, b) \mapsto a \cdot b$$

and the subtraction

$$\mathbb{Z} \times \mathbb{Z} \longrightarrow \mathbb{Z} \quad (a, b) \mapsto a - b$$

are internal composition laws.

2. Let E be a non-empty set. On $\mathcal{P}(E)$, the union defined from $\mathcal{P}(E)^2$ to $\mathcal{P}(E)$ by $(A, B) \mapsto A \cup B$ is an internal composition law because $A \cup B \in \mathcal{P}(E)$.

We have another example of an internal composition law.

Example

We equip $\mathbb{R}$ with the internal composition law $*$ defined by :

$$\forall x, y \in \mathbb{R} : \quad x * y = xy + \left(x^2 - 1\right)\left(y^2 - 1\right)$$

We have $xy \in \mathbb{R}$, $(x^2 - 1) \in \mathbb{R}$ and $(y^2 - 1) \in \mathbb{R}$. Therefore, $x * y \in \mathbb{R}$.

4.1.2 Subsets closed under a composition law

Definition

Let E be a non-empty set equipped with an internal composition law $*$, and let A be a non-empty subset of E, A is said to be closed under the law $*$ if

$$\forall (x, y) \in A \times A : \quad x * y \in A.$$

Example

The set $\mathbb{Z}$ equipped with the ICL (addition) $+$. The set of even integer $2\mathbb{Z}$ which is subset of $\mathbb{Z}$ is closed under the addition.

4.1.3 Properties of an Internal Composition Law

Commutativity and Associativity

Definition

Let E be a non-empty set equipped with $*$ an Internal Composition Law. We say that:

- $*$ is commutative if for all $x, y \in E$: $x * y = y * x$.
- $*$ is associative if for all $x, y, z \in E$: $(x * y) * z = x * (y * z)$.

Example

- The set $\mathbb{N}$ equipped with the addition $+$, $+$ is associative:

$$\forall a, b, c \in \mathbb{N} : \ (a + b) + c = a + (b + c).$$

- On $\mathbb{R}$, the multiplication $\cdot$ is commutative:

$$\forall a, b \in \mathbb{R} : \ a \cdot b = b \cdot a.$$

Neutral Element(Identity Element)

Definition

Let E be a non-empty set equipped with $*$ an Internal Composition Law. We say that an element $e \in E$ is a neutral element (or identity element) for $*$ if for all $x \in E$, we have

$$x * e = e * x = x$$

Proposition

Let E be a non-empty set equipped with $*$ an Internal Composition Law. If $*$ has a neutral element (i.e. there exists a neutral element for $*$) then it is unique.

Proof. We suppose that $*$ has two neutral elements e and e' in E, then we have respectively $e' * e = e * e' = e'$ and $e * e' = e' * e = e$ therefore $e = e'$. □

Example
On $\mathbb{Z}$ equipped with the addition $+$, the neutral element for $+$ is 0, we have:

$$\forall a \in \mathbb{Z}: \; a + 0 = 0 + a = a.$$

Inverse Element

Definition

Let E be a non-empty set equipped with $*$ an internal composition law that has a neutral element e in E and let x be an element of E. We say that x has an inverse element for the law $*$ if there exists $y \in E$ such that

$$x * y = y * x = e$$

The inverse of an element x is usually denoted by x^{-1} or $-x$. These notations are symbolic and do not represent exponentiation by -1 or multiplication of x by -1.

Example
On $\mathbb{Z}$ equipped with $+$, the inverse of every element $a \in \mathbb{Z}$ is $-a$, we have:

$$\forall a \in \mathbb{Z}; \; a + (-a) = (-a) + a = 0.$$

Distributivity

Definition

Let E be a non-empty set equipped with two internal composition laws

$$* : E \times E \to E, \qquad \diamond : E \times E \to E.$$

We say that $*$ is **left-distributive** over $\diamond$ if, for all $x, y, z \in E$,

$$x * (y \diamond z) = (x * y) \diamond (x * z).$$

We say that $*$ is **right-distributive** over $\diamond$ if, for all $x, y, z \in E$,

$$(y \diamond z) * x = (y * x) \diamond (z * x).$$

If both properties hold, we say that $*$ is **distributive** over $\diamond$.

Example
In the set of real numbers $\mathbb{R}$, the multiplication $\cdot$ is distributive over addition $+$:

$$\forall a, b, c \in \mathbb{R}: \; a \cdot (b + c) = (a \cdot b) + (a \cdot c). \quad \text{(left distributivity)}$$

$$\forall a, b, c \in \mathbb{R}: \; (b + c) \cdot a = (b \cdot a) + (c \cdot a). \quad \text{(right distributivity)}$$

4.2 Groups

4.2.1 Definitions

Definition

A **group** $(G, *)$ is a non-empty set G equipped with an internal composition law $*$ verifying the following properties:

1. For all $x, y, z \in G$: $(x * y) * z = x * (y * z)$ (**The law is associative**)
2. There exists $e \in G, \forall x \in G: \ x * e = x$ and $e * x = x$ (e is the **neutral element**)
3. For all $x \in G$, there exists $y \in G$ such that $x * y = y * x = e$ (y is the **inverse** of x)

Definition

Let E be a non-empty set equipped with an internal composition law $*$. An abelian **group** or commutative group $(G, *)$ is a group with the law $*$ is commutative.

Examples

1. $(\mathbb{Z}, +)$ is a commutative (abelian) group.
2. $(\mathbb{R}, +)$ is a commutative (abelian) group.
3. $(\mathbb{C}, +)$ is a commutative (abelian) group.

4.2.2 Subgroups

Definition

Let $(G, *)$ be a group and let H be a non-empty subset of G. We say that H is a **subgroup** of G if H is closed under the group law and forms a group with the restriction of the law on H, that is:

- for all $x, y \in H$, we have $x * y \in H$,
- the neutral element of G belongs to H,
- for every $x \in H$, its inverse x^{-1} belongs to H.

In this case, we write $H \leq G$.

*Note that a subgroup H is also a group $(H, *)$ with the law **restricted from** G.*

Proposition

Let $(G, *)$ be a group and let H be a non-empty subset of G. Then H is a subgroup of G *if and only if* for all $x, y \in H$, we have

$$x * y^{-1} \in H.$$

Examples of Subgroups

1. $(2\mathbb{Z}, +)$ is a subgroup of $(\mathbb{Z}, +)$.
2. $(\mathbb{Z}, +)$ is a subgroup of $(\mathbb{Q}, +)$.
3. $(\mathbb{R}_+^*, \cdot)$ (the positive real numbers under multiplication) is a subgroup of $(\mathbb{R}^*, \cdot)$.

Example of a Non-Subgroup

- $(\mathbb{N}, +)$ is not a subgroup of $(\mathbb{Z}, +)$ because it is not closed under inverses.

4.2.3 Examples of Groups

The group $\mathbb{Z}/n\mathbb{Z}$

Let $n \in \{1, 2, 3, \ldots\}$. We define a binary relation $\mathcal{R}$ on $\mathbb{Z}$ by

$$a \,\mathcal{R}\, b \quad \Longleftrightarrow \quad a \equiv b \,[n] \quad \Longleftrightarrow \quad \exists\, k \in \mathbb{Z} \text{ such that } a - b = kn.$$

We prove that $\mathcal{R}$ is an equivalence relation.

1. Reflexivity

To prove: $\forall a \in \mathbb{Z},\ a\mathcal{R}a$.
Proof: Let $a \in \mathbb{Z}$. We have

$$a - a = 0 = 0 \cdot n,$$

and since $0 \in \mathbb{Z}$, there exists $k = 0$ such that

$$a - a = kn.$$

Hence, $a\mathcal{R}a$. Therefore, $\mathcal{R}$ is reflexive. □

2. Symmetry

To prove: $\forall a, b \in \mathbb{Z},\ a\mathcal{R}b \Rightarrow b\mathcal{R}a$.
Proof: Assume that $a\mathcal{R}b$. Then there exists $k \in \mathbb{Z}$ such that

$$a - b = kn.$$

Multiplying both sides by -1, we obtain

$$b - a = (-k)n,$$

and since $-k \in \mathbb{Z}$, it follows that $b\mathcal{R}a$. Therefore, $\mathcal{R}$ is symmetric. □

3. Transitivity

To prove: $\forall a, b, c \in \mathbb{Z},\ a\mathcal{R}b$ and $b\mathcal{R}c \Rightarrow a\mathcal{R}c$.

Proof: Assume that $a\mathcal{R}b$ and $b\mathcal{R}c$. Then there exist $k_1, k_2 \in \mathbb{Z}$ such that

$$a - b = k_1 n \quad \text{and} \quad b - c = k_2 n.$$

Adding these two equalities, we get

$$a - c = (k_1 + k_2)n.$$

Since $k_1 + k_2 \in \mathbb{Z}$, it follows that $a\mathcal{R}c$. Therefore, $\mathcal{R}$ is transitive. □

Since $\mathcal{R}$ is reflexive, symmetric, and transitive, it is an equivalence relation on $\mathbb{Z}$.

Equivalence classes

For each $a \in \mathbb{Z}$, the equivalence class of a is

$$\overline{a} = \{\, b \in \mathbb{Z} \mid a\mathcal{R}b \,\} = \{\, b \in \mathbb{Z} \mid \exists\, k \in \mathbb{Z} : a - b = kn \,\}.$$

Equivalently, we can write

$$\overline{a} = \{\, a + tn \mid t \in \mathbb{Z} \,\}.$$

We have the following property for all $a, b \in \mathbb{Z}$:

$$\overline{a} = \overline{b} \quad \Longleftrightarrow \quad a \equiv b\ [n].$$

Proof:

1. ($\Rightarrow$) We suppose that $\overline{a} = \overline{b}$. Since $a \in \overline{a}$, we get $a \in \overline{b}$, so $a = b + tn$ with $t \in \mathbb{Z}$. Thus $a \equiv b\ [n]$.

2. ($\Leftarrow$) We suppose $a \equiv b\ [n]$. Then $a = b + tn$ with $t \in \mathbb{Z}$. Let $x \in \overline{a}$, so $x = a + ln$ with $l \in \mathbb{Z}$. Then
$$x = (b + tn) + ln = b + (t + l)n.$$
then, $x \in \overline{b}$. Thus, $\overline{a} \subseteq \overline{b}$.
We supposed that $a \equiv b\ [n]$ by the symmetry property of the equivalence relation we have $b \equiv a\ [n]$. Then $b = a + tn$ with $t \in \mathbb{Z}$. Let $x \in \overline{b}$, so $x = b + ln$ with $l \in \mathbb{Z}$. Then
$$x = (b + tn) + ln = b + (t + l)n,$$
then, $x \in \overline{a}$. Thus, $\overline{b} \subseteq \overline{a}$.
Therefore, we get both inclusions then the equality hold, $\overline{a} = \overline{b}$.

The quotient set is

$$\mathbb{Z}/n\mathbb{Z} = \{\, \overline{a} \ \mid\ a \in \mathbb{Z} \,\} = \{\, \overline{0}, \overline{1}, \ldots, \overline{n-1} \,\}.$$

because for example we have: $\overline{0} = \{\, tn \mid t \in \mathbb{Z} \,\} = \overline{tn}$, for all $t \in \mathbb{Z}$. Also $\overline{1} = \{\, 1 + tn \mid t \in \mathbb{Z} \,\} = \overline{1 + tn}$, for all $t \in \mathbb{Z}$. The same reasoning applies to all other classes.

Since many equivalence classes are equal, we do not list them all inside the quotient set. We choose only one representative from each distinct equivalence class.

We define the operation $\oplus$ on $\mathbb{Z}/n\mathbb{Z}$ by

$$\overline{a} \oplus \overline{b} = \overline{a + b}.$$

Then $(\mathbb{Z}/n\mathbb{Z}, \oplus)$ is a commutative group. Indeed, we have

- $\oplus$ is an internal composition law on $\mathbb{Z}/n\mathbb{Z}$.

$$\forall\, \overline{a}, \overline{b} \in \mathbb{Z}/n\mathbb{Z}, \quad \overline{a} \oplus \overline{b} \in \mathbb{Z}/n\mathbb{Z}.$$

  Proof: Let $\overline{a}, \overline{b} \in \mathbb{Z}/n\mathbb{Z}$. By definition $\overline{a} \oplus \overline{b} = \overline{a+b}$, and $\overline{a+b}$ is one of the classes in $\mathbb{Z}/n\mathbb{Z}$, because $\forall a, b \in \mathbb{Z} : \; a + b \in \mathbb{Z}$.

- **Associativity:**

$$\forall\, \overline{a}, \overline{b}, \overline{c} \in \mathbb{Z}/n\mathbb{Z}, \quad (\overline{a} \oplus \overline{b}) \oplus \overline{c} = \overline{a} \oplus (\overline{b} \oplus \overline{c}).$$

  Proof: Let $\overline{a}, \overline{b}, \overline{c} \in \mathbb{Z}$, for representatives $a, b, c \in \mathbb{Z}$, we have

$$(\overline{a} \oplus \overline{b}) \oplus \overline{c} = \overline{a+b} \oplus \overline{c} = \overline{(a+b)+c},$$

  and

$$\overline{(a+b)+c} = \overline{a+(b+c)} = \overline{a} \oplus \overline{b+c} = \overline{a} \oplus (\overline{b} \oplus \overline{c}),$$

  since $(a+b)+c = a+(b+c)$ in $\mathbb{Z}$. Therefore, $(\overline{a} \oplus \overline{b}) \oplus \overline{c} = \overline{a} \oplus (\overline{b} \oplus \overline{c})$.

- **Identity:**

$$\exists\, E \in \mathbb{Z}/n\mathbb{Z}, \; \forall\, \overline{a} \in \mathbb{Z}/n\mathbb{Z} : \quad \overline{a} \oplus E = \overline{a}.$$

  Proof: We take $E = \overline{0}$. For any $\overline{a} \in \mathbb{Z}/n\mathbb{Z}$,

$$\overline{a} \oplus \overline{0} = \overline{a+0} = \overline{a},$$

  so $\overline{0}$ satisfies the requirement.

- **Inverse:**

$$\forall\, \overline{a} \in \mathbb{Z}/n\mathbb{Z}, \; \exists\, \overline{b} \in \mathbb{Z}/n\mathbb{Z} \text{ such that } \overline{a} \oplus \overline{b} = \overline{0}.$$

  Proof: For given $\overline{a}$, we take $\overline{b} = \overline{-a}$. Then

$$\overline{a} \oplus \overline{-a} = \overline{a+(-a)} = \overline{0},$$

  so an inverse exists.

- **Commutativity:**

$$\forall\, \overline{a}, \overline{b} \in \mathbb{Z}/n\mathbb{Z}, \quad \overline{a} \oplus \overline{b} = \overline{b} \oplus \overline{a}.$$

  Proof: Let $\overline{a}, \overline{b} \in \mathbb{Z}$, for representatives $a, b \in \mathbb{Z}$, we have,

$$\overline{a} \oplus \overline{b} = \overline{a+b} = \overline{b+a} = \overline{b} \oplus \overline{a},$$

  since $a + b = b + a$ in $\mathbb{Z}$.

Thus $(\mathbb{Z}/n\mathbb{Z}, \oplus)$ is an abelian group.

Permutation Group

Proposition

Let $n \geq 2$. The set of bijections from $\{1, 2, \ldots, n\}$ to itself, equipped with $\circ$, is a group. This group is denoted by $(S_n, \circ)$ and its elements are called permutations.
Furtheremore, the number of elements of S_n is $|S_n| = n!$.

Proof. Let S_n be the set of all functions $f : \{1, \ldots, n\} \to \{1, \ldots, n\}$ that are bijections.

1. $\circ$ is an internal composition law on S_n. For all $f, g \in S_n$, the composition $g \circ f$ is again a bijection from $\{1, \ldots, n\}$ to itself. Therefore $g \circ f \in S_n$.

2. **Associativity.** For all $f, g, h \in S_n$, the composition of functions satisfies
$$h \circ (g \circ f) = (h \circ g) \circ f.$$

3. **Identity.** The map
$$Id_{\{1,\ldots,n\}} : i \mapsto i$$
is a bijection and satisfies
$$Id_{\{1,\ldots,n\}} \circ f = f \qquad \text{and} \qquad f \circ Id_{\{1,\ldots,n\}} = f$$
for all $f \in S_n$. Hence $Id_{\{1,\ldots,n\}}$ is the neutral element of $(S_n, \circ)$.

4. **Inverse.** For every bijection $f \in S_n$, there exists a function f^{-1} such that
$$f \circ f^{-1} = Id_{\{1,\ldots,n\}} \quad \text{and} \quad f^{-1} \circ f = Id_{\{1,\ldots,n\}}.$$
The inverse of a function f is obtained by reversing their preimages and images.

The Group S_3

The set S_3 consists of all bijections of $\{1, 2, 3\}$. Since $3! = 6$, the group $(S_3, \circ)$ has 6 elements.

We write each permutation as a function and also in two-line notation using parentheses:

$$f_{\text{id}} : 1 \mapsto 1,\ 2 \mapsto 2,\ 3 \mapsto 3 \qquad \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}$$

$$f_{\tau_1} : 1 \mapsto 1,\ 2 \mapsto 3,\ 3 \mapsto 2 \qquad \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}$$

$$f_{\tau_2} : 1 \mapsto 3,\ 2 \mapsto 2,\ 3 \mapsto 1 \qquad \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}$$

$$f_{\tau_3} : 1 \mapsto 2,\ 2 \mapsto 1,\ 3 \mapsto 3 \qquad \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}$$

$$f_{\sigma} : 1 \mapsto 2,\ 2 \mapsto 3,\ 3 \mapsto 1 \qquad \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$$

$$f_{\sigma^{-1}} : 1 \mapsto 3,\ 2 \mapsto 1,\ 3 \mapsto 2 \qquad \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}$$

Thus
$$S_3 = \{f_{\text{id}}, f_{\tau_1}, f_{\tau_2}, f_{\tau_3}, f_{\sigma}, f_{\sigma^{-1}}\}.$$

How to find the inverse of a permutation

To obtain the inverse of a permutation written as

$$\begin{pmatrix} 1 & 2 & 3 \\ a & b & c \end{pmatrix},$$

we reverse the mapping, for example: a the image of 1 becomes the preimage and 1 becomes the image, etc.

Example:

$$f_\sigma = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}.$$

Since

$$1 \mapsto 2, \quad 2 \mapsto 3, \quad 3 \mapsto 1,$$

we reverse the arrows, we get

$$2 \mapsto 1, \quad 3 \mapsto 2, \quad 1 \mapsto 3,$$

$$f_{\sigma^{-1}} = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}.$$

Composition Example

Let us compute $f_{\tau_1} \circ f_\sigma$:

$$f_\sigma : 1 \mapsto 2,\ 2 \mapsto 3,\ 3 \mapsto 1$$
$$f_{\tau_1} : 1 \mapsto 1,\ 2 \mapsto 3,\ 3 \mapsto 2.$$

Then

$$(f_{\tau_1} \circ f_\sigma)(1) = f_{\tau_1}(2) = 3, \qquad (f_{\tau_1} \circ f_\sigma)(2) = f_{\tau_1}(3) = 2, \qquad (f_{\tau_1} \circ f_\sigma)(3) = f_{\tau_1}(1) = 1.$$

Thus

$$f_{\tau_1} \circ f_\sigma = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} = f_{\tau_2}.$$

Similarly,

$$f_\sigma \circ f_{\tau_1} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix} = f_{\tau_3}.$$

Hence

$$f_{\tau_1} \circ f_\sigma \neq f_\sigma \circ f_{\tau_1},$$

so $(S_3, \circ)$ is not commutative.

Composition Table of $(S_3, \circ)$

$\circ$	f_{id}	f_{τ_1}	f_{τ_2}	f_{τ_3}	f_σ	f_σ^{-1}
f_{id}	f_{id}	f_{τ_1}	f_{τ_2}	f_{τ_3}	f_σ	f_σ^{-1}
f_{τ_1}	f_{τ_1}	f_{id}	f_σ	f_σ^{-1}	f_{τ_2}	f_{τ_3}
f_{τ_2}	f_{τ_2}	f_σ^{-1}	f_{id}	f_σ	f_{τ_3}	f_{τ_1}
f_{τ_3}	f_{τ_3}	f_σ	f_σ^{-1}	f_{id}	f_{τ_1}	f_{τ_2}
f_σ	f_σ	f_{τ_3}	f_{τ_1}	f_{τ_2}	f_σ^{-1}	f_{id}
f_σ^{-1}	f_σ^{-1}	f_{τ_2}	f_{τ_3}	f_{τ_1}	f_{id}	f_σ

This table summarises all possible compositions in the group $(S_3, \circ)$.

4.2.4 Group Homomorphism

Definition

Let $(G, \perp)$ and $(H, *)$ be two groups. A map $f : G \to H$ is called a **group homomorphism** if, for all $x, y \in G$,

$$f(x \perp y) = f(x) * f(y).$$

In other words, f preserves the group operation.

Furtheremore, if f is a group homomorphism then we have:

$$f(e_G) = e_H$$

where e_G and e_H are the neutral elements of $\perp$ and $*$ respectively.
and it verifies the following property

$$f(x^{-1}) = (f(x))^{-1}$$

where x^{-1} noted the inverse of x and $(f(x))^{-1}$ noted the inverse of $f(x)$.

Example

Consider the map $f : (\mathbb{R}, +) \to (\mathbb{R}, +)$ defined by

$$f(x) = 2x, \qquad \text{for all } x \in \mathbb{R}.$$

This map is a group homomorphism from $(\mathbb{R}, +)$ to $(\mathbb{R}, +)$ because

$$\text{for all } x, y \in \mathbb{R} : \; f(x + y) = 2(x + y) = 2x + 2y = f(x) + f(y).$$

4.2.5 Group Isomorphism

Definition

Let $(G, \perp)$ and $(H, *)$ be two groups. A map $f : G \to H$ is called a **group isomorphism** if:

- f is a group homomorphism, and
- f is bijective.

In this case, the two groups $(G, \perp)$ and $(H, *)$ are said to be **isomorphic**, and we write

$$(G, \perp) \simeq (H, *).$$

Example

The exponential map $\exp : \mathbb{R} \to \mathbb{R}_+^*$ is a group homomorphism from $(\mathbb{R}, +)$ to $(\mathbb{R}_+^*, \cdot)$, because

$$\exp(x + y) = \exp(x)\exp(y), \qquad \text{for all } x, y \in \mathbb{R}.$$

Since this map is also a bijection, it follows that exp is a group isomorphism.

4.3 Rings

4.3.1 Definition

Definition

A ring is a set R equipped with two internal composition laws

$$\oplus : R \times R \to R \qquad \text{and} \qquad \odot : R \times R \to R,$$

such that:

1. $(R, \oplus)$ is an abelian group:
   - For all $x, y, z \in R$ $x \oplus (y \oplus z) = (x \oplus y) \oplus z$.
   - $\exists e \in R$, $\forall x \in R$ such that $x \oplus e = e \oplus x = x$. We note the neutral element e by 0_R.
   - For every $x \in R$ there exists $y \in R$ such that $x \oplus y = y \oplus x = 0_R$;
   - $\forall x, y \in R :\ x \oplus y = y \oplus x$.
2. $\odot$ is associative:
   $$\forall x, y, z \in R : (x \odot y) \odot z = x \odot (y \odot z).$$
3. $\odot$ distributes over $\oplus$: For all $x, y, z \in R$ we have
   $$x \odot (y \oplus z) = (x \odot y) \oplus (x \odot z),$$
   $$(x \oplus y) \odot z = (x \odot z) \oplus (y \odot z).$$

Definition

A ring $(R, \oplus, \odot)$ is said to be unital ring if $\odot$ has an identity element in R noted 1_R.

Remark

In some references, the authors add that the property: $\odot$ has an identity element in R must be in the definition of a ring.

Definition

A ring $(R, \oplus, \odot)$ is said to be a commutative ring if $\odot$ is commutative.

Examples of Commutative Unitary Rings

1. $(\mathbb{Z}, +, \cdot)$ is a commutative unital ring.
2. $(\mathbb{R}, +, \cdot)$ is a commutative unital ring.
3. $(\mathbb{C}, +, \cdot)$ is a commutative unital ring.

Remark

In this course, we use the following notations in a ring $(R, \oplus, \odot)$:

- For any $x \in R$, $-x$ denotes the inverse of x with respect to $\oplus$.
- For any $x \in R$, x^{-1} (when it exists) denotes the inverse of x with respect to $\odot$.

4.3.2 Subrings

Definition

Let $(R, \oplus, \odot)$ be a ring, and let A be a subset of R. The set A, equipped with the restrictions of the internal composition laws $\oplus$ and $\odot$ to A, is said to be a subring of $(R, \oplus, \odot)$ if it keeps the same ring structure under these induced internal composition laws.

Proposition

Let $(R, \oplus, \odot)$ be a ring. A subset $A \subseteq R$, $(A \neq \emptyset)$ equipped with the restrictions of the internal composition laws $\oplus$ and $\odot$ is a **subring** of $(R, \oplus, \odot)$ if and only if:

- $\forall x, y \in A: \; x \oplus (-y) \in A$;
- $\forall x, y \in A: \; x \odot y \in A$.

Examples of Subrings

1. $(\mathbb{Z}, +, \cdot)$ is a subring of $(\mathbb{Q}, +, \cdot)$.
2. $(\mathbb{Q}, +, \cdot)$ is a subring of $(\mathbb{R}, +, \cdot)$.
3. $(2\mathbb{Z}, +, \cdot)$ is a subring of $(\mathbb{Z}, +, \cdot)$.

4.3.3 Calculation Rules in a Ring

Let $(R, \oplus, \odot)$ be a ring, and let $x, y \in R$. Then:

- For any $x \in R$,
$$x \odot 0_R = 0_R \odot x = 0_R$$
- For any $x, y \in R$,
$$x \odot (-y) = -(x \odot y)$$
$$(-x) \odot y = -(x \odot y)$$
- For any $x, y \in R$,
$$(-x) \odot (-y) = x \odot y$$
- For all $n \in \mathbb{Z}$, $x, y \in R$:
$$n(x \odot y) = (nx) \odot y = x \odot (ny),$$
where nx denotes the repeated use of the law $\oplus$ (and not the law $\odot$).

Remark

Let $(R, \oplus, \odot)$ be a ring, and let $x \in R$ and $n \in \mathbb{Z}$. We define nx as follows:

- If $n > 0$, then
$$nx = \underbrace{x \oplus x \oplus \cdots \oplus x}_{n \text{ times}}.$$
- If $n = 0$, then
$$0x = 0_R.$$
- If $n < 0$, write $n = -m$ with $m > 0$. Then
$$nx = (-m)x = -(mx) = -\Big(\underbrace{x \oplus x \oplus \cdots \oplus x}_{m \text{ times}}\Big),$$
where the minus sign denotes the inverse with respect to $\oplus$.

Additional rules in a unitary ring.

Assume now that R is a unital ring with neutral element 1_R for $\odot$, and that x and y commute (i.e., $x \odot y = y \odot x$). Then:

- $$(x \oplus y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k \odot y^{n-k};$$
- $$x^n \oplus (-y^n) = (x \oplus (-y)) \odot \sum_{k=0}^{n-1} x^k \odot y^{n-1-k}.$$

Remark

- The symbol $\binom{n}{k}$ denotes the binomial coefficient:

$$\binom{n}{k} = \frac{n!}{k!\,(n-k)!}.$$

- In the above formulas, the symbol $\sum$ does not denote the usual addition. It represents an iterated application of the operation $\oplus$. More precisely,

$$\sum_{k=0}^{n} a_k \;=\; a_0 \oplus a_1 \oplus \cdots \oplus a_n.$$

- The binomial coefficient $\binom{n}{k}$ denotes repeated $\oplus$, that is,

$$\binom{n}{k} a = \underbrace{a \oplus a \oplus \cdots \oplus a}_{\binom{n}{k} \text{ times}}.$$

- For any $x \in R$:

$$x^0 = 1_R, \qquad x^n = \underbrace{x \odot x \odot \cdots \odot x}_{n \text{ times}} \text{ for } n \geq 1.$$

  If R is not unitary, only the powers x^n for $n \geq 1$ are defined. Thus, the usual exponent notation does not refer to the usual multiplication of real numbers, but to the operation $\odot$ of the ring.

4.3.4 Invertible Elements, Zero Divisors

Definition

Let $(R, \oplus, \odot)$ be a unital ring, with neutral element 1_R for $\odot$. An element $x \in R$ is called **invertible** (or a *unit*) if there exists $y \in R$ such that

$$x \odot y = y \odot x = 1_R.$$

Such an element y is unique and is denoted by x^{-1}.
The set of invertible elements of R forms a group for $\odot$, called the **group of units**, denoted by $R^\times$.

Definition

Let $(R, \oplus, \odot)$ be a ring. A non-zero element $x \in R$ is a **zero divisor** if there exists a non-zero $y \in R$ such that

$$x \odot y = 0_R.$$

Remark

In a ring $(R, \oplus, \odot)$, the term *non-zero element* does not mean an element different from the number 0 in $\mathbb{Z}$. It means an element different from the *neutral element* of the first law $\oplus$. More precisely, if 0_R denotes the neutral element of $(R, \oplus)$, then

$$a \in R \text{ is non-zero} \quad \Longleftrightarrow \quad a \neq 0_R.$$

Example 1: Ring with zero divisors

Let

$$R_1 = \mathbb{Z}/4\mathbb{Z}.$$

We denote by $\overline{a}$ the equivalence class of $a \in R_1$.

The operations $\oplus$ and $\odot$ are defined by

$$\overline{a} \oplus \overline{b} = \overline{a+b}, \qquad \overline{a} \odot \overline{b} = \overline{ab}.$$

We take the non-zero element (different to the neutral element of the first law $\oplus$ that is $\overline{0}$) $\overline{2} \in R_1$ ($\overline{2} \neq \overline{0}$). We have

$$\overline{2} \odot \overline{2} = \overline{4} = \overline{0}.$$

Thus, $\overline{2} \neq \overline{0}$ and

$$\overline{2} \odot \overline{2} = \overline{0}.$$

Hence, $\overline{2}$ is a *zero divisor* in $(R_1, \oplus, \odot)$.

Example 2: Ring without zero divisors

Let

$$R_2 = \mathbb{Z}.$$

$\mathbb{Z}$ with the usual addition and multiplication .

We assume that $a, b \in \mathbb{Z}$ satisfy

$$a \cdot b = 0.$$

Then either $a = 0$ or $b = 0$. Therefore, there do not exist non-zero elements $a, b \in \mathbb{Z}$ such that

$$a \cdot b = 0.$$

Hence, $(\mathbb{Z}, \oplus, \odot)$ has *no zero divisors*.

4.3.5 Ring Homomorphism

Definition

Let $(R, \oplus, \odot)$ and $(S, \perp, *)$ be rings. A map $f : R \to S$ is called a **ring homomorphism** if:

- for all $x, y \in R : f(x \oplus y) = f(x) \perp f(y)$,
- for all $x, y \in R : f(x \odot y) = f(x) * f(y)$.

If f is bijective, it is called a **ring isomorphism**.

Definition

Let $(R, \oplus, \odot)$ and $(S, \perp, *)$ be unital rings, with neutral elements 1_R and 1_S for $\odot$. A map $f : R \to S$ is a **unital ring homomorphism** if:

- $f(1_R) = 1_S$;
- $f(x \oplus y) = f(x) \perp f(y)$ for all $x, y \in R$;
- $f(x \odot y) = f(x) * f(y)$ for all $x, y \in R$.

If f is bijective, it is a **unital ring isomorphism**.

Remark

For any ring homomorphism $f : R \to S$, one has:

$$f(0_R) = 0_S, \qquad f(nx) = n\, f(x) \quad \text{for all } n \in \mathbb{Z},\ x \in R,$$

4.3.6 Ideals

Definition

Let $(R, \oplus, \odot)$ be a commutative ring. A nonempty subset $I \subseteq R$ is called an ideal of R if it satisfies:

1. $(I, \oplus)$ is a subgroup of $(R, \oplus)$.
2. For all $x \in I$ and $a \in R$, the composition $a \odot x$ belongs to I. Since R is commutative, we also have $x \odot a \in I$.

Examples

1. For every $n \in \mathbb{Z}$, the set $n\mathbb{Z}$ of multiples of n is an ideal of the ring $(\mathbb{Z}, +, \cdot)$.

4.4 Fields

4.4.1 Definitions

Definition

A field is a commutative unital ring $(F, \oplus, \odot)$ in which every nonzero element has an inverse for $\odot$, that is,

$$\forall x \in F \setminus \{0_F\},\ \exists\, y \in F \text{ with } x \odot y = 1_F.$$

We denote by 0_F the neutral element of the first law $\oplus$, and by 1_F the neutral element of the second law $\odot$.

1. The field of real numbers, $(\mathbb{R}, +, .)$.
2. The field of complex numbers, $(\mathbb{C}, +, .)$.

4.4.2 Finite Fields

Definition

Let $(F, \oplus, \odot)$ be a field is said to be finite if the set F is finite.

Definition

On the quotient set

$$\mathbb{Z}/n\mathbb{Z} = \{\, \overline{a} \mid a \in \mathbb{Z} \,\},$$

we define two operations:

- **First law :**

$$\forall a, b \in \mathbb{Z}, \qquad \overline{a} \oplus \overline{b} := \overline{a+b}.$$

  Its neutral element is

$$0_{\mathbb{Z}/n\mathbb{Z}} = \overline{0}.$$

- **Second law :**

$$\forall a, b \in \mathbb{Z}, \qquad \overline{a} \odot \overline{b} := \overline{a \cdot b}.$$

  Its neutral element is

$$1_{\mathbb{Z}/n\mathbb{Z}} = \overline{1}.$$

Theorem

Let $p \geq 1$ be an integer. The finite ring $(\mathbb{Z}/p\mathbb{Z}, \oplus, \odot)$ is a commutative unitary ring. Moreover, it is a field if and only if p is a prime number. In this case, this field is denoted by $\mathbb{F}_p$.

Remark

When p is prime, the second law $\odot$ gives an inverse to every nonzero class $\overline{a}$.

Example

- The finite field $(\mathbb{Z}/2\mathbb{Z}, \oplus, \odot)$, where 2 is a prime number.

Test Your Chapter Mastery

Exercise 17: True or False ?

1. Answer True or False. Justify your answers.
   (a) Let H be the set of odd integers, $(H, +)$ is a subgroup of $(\mathbb{Z}, +)$.

(b) $H = [-1, +\infty[$ equipped with the addition is not a subgroup of $G = (\mathbb{R}, +)$.

(c) Let $H = \{f \in G; f(x) = x\}$, $(H, \circ)$ is not a subgroup of $(G, , \circ)$, where $G = \{$ bijections from E to $E\}$.

(d) Let

$$T = \{\, 3k \mid k \in \mathbb{Z} \,\} \subseteq \mathbb{Z}.$$

$(T, +, \cdot)$ is a subring of $(\mathbb{Z}, +, \cdot)$.

2. Let the following function:

$$\varphi : G \to H$$

φ is a group homomorphism? a group isomomorphism ? in the following cases?

- $G = (\mathbb{R}^*, \cdot)\,, H = (\mathbb{R}^*, \cdot)\,, \varphi$ defined by: $\varphi(x) = |x|$.
- $G = (\mathbb{R}^*, \cdot)\,, \quad H = (\mathbb{R}^*, \cdot)\,, \varphi$ defined by: $\varphi(x) = 2x$

Exercise 18: Group, Ring and Field

1)We consider the set $G = \mathbb{R}$, and the law $\perp$ defined as

$$\forall x, y \in G : \qquad x \perp y = x + y - 1$$

1. Prove that $\perp$ is an internal composition law on G.
2. Prove that $(G, \perp)$ is an abelian group.

2) On the same set $G = \mathbb{R}$, we consider a second law $*$ defined by

$$\forall x, y \in G : \qquad x * y = xy - x - y + 2.$$

With the two operations $\perp$ and $*$ defined above on $G = \mathbb{R}$:

1. Prove that the structure $(G, \perp, *)$ is a ring.
2. If it is a ring, determine whether it is commutative.
3. Determine whether $(G, \perp, *)$ is a field.

Exercise 19: Ring Homomorphism

On $\mathbb{R}$ we define two internal composition laws:

$$a \oplus b = a + b - 1, \qquad a \odot b = ab - a - b + 2.$$

1. We consider the function

$$f : (\mathbb{R}, \oplus, \odot) \longrightarrow (\mathbb{R}, +, \cdot), \qquad f(x) = x - 1.$$

Show that f is a ring homomorphism.

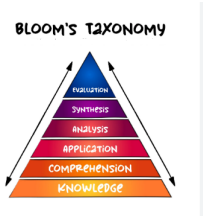
2. Is f a unital ring homomorphism?

5

Polynomial Rings

Specific Objectives

By the end of this chapter, students will be able to:



1. Define polynomials and their degree.
2. Construct the polynomial ring over a ring or a field..
3. Perform arithmetic operations on polynomials.
4. Apply Euclidean division to polynomials.
5. Determine the GCD and LCM of polynomials.
6. Factor polynomials into irreducible factors.

5.1 Polynomials and Degree

Definition

A polynomial P with coefficients in a commutative unital ring K has an expression of the form:

$$P(X) = a_0 + a_1X + a_2X^2 + \cdots + a_{n-1}X^{n-1} + a_nX^n, \quad a_i \in K, n \in \mathbb{N}.$$

$a_i \in K$ are called the coefficients of the polynomial.

Remark

The set K of coefficients may be a field.

Definition

The set of all polynomials with coefficients in K noted $K[X]$.

Definition

P is called the zero polynomial if all the coefficients $a_i \in K$ are zero, i.e. $P(X) = 0, \ \forall X$.

Definition

The **degree** $\deg(P)$ is the highest power of X with a non-zero coefficient.

Example

Let $P(X) = 2X^3 + X - 1$. Then

$$\deg(P) = 3, \quad \text{coefficients: } 2, 0, 1, -1 \in \mathbb{R}$$

Let $Q(X) = X^n + 1$ with $deg(Q) = n$.

5.2 Construction of the Polynomial Ring

Let K be a commutative ring with identity and let $P, Q \in K[X]$.

Addition

Polynomial addition is defined as follows:

$$(P + Q)(X) = \sum_{i=0}^{n} (a_i + b_i) X^i.$$

Multiplication

Let

$$P(X) = \sum_{i=0}^{n} a_i X^i \quad \text{and} \quad Q(X) = \sum_{j=0}^{m} b_j X^j.$$

Their product is

$$(P \cdot Q)(X) = \sum_{k=0}^{n+m} c_k X^k, \quad \text{where} \quad c_k = \sum_{i=0}^{k} a_i b_{k-i}.$$

Example
Given: $P(x) = 2 + 3x$ and $Q(x) = 1 - 5x$
The product is $S(x) = \sum_{k=0}^{1+1} c_k x^k$, where $c_k = \sum_{i=0}^{k} a_i b_{k-i}$.

Coefficient c_0 (for x^0):
$c_0 = a_0 b_0 = (2)(1) = 2$

Coefficient c_1 (for x^1):
$c_1 = a_0 b_1 + a_1 b_0 = (2)(-5) + (3)(1) = -10 + 3 = -7$

Coefficient c_2 (for x^2):
$c_2 = a_0 b_2 + a_1 b_1 + a_2 b_0$
Since $a_2 = 0$ and $b_2 = 0$:
$c_2 = (2)(0) + (3)(-5) + (0)(1) = 0 - 15 + 0 = -15$

The resulting product is:
$P(x) \cdot Q(x) = 2 - 7x - 15x^2$

Definition

The set of all polynomials with coefficients in K (K is a commutative unital ring or a field) noted $K[X]$, equipped with addition and multiplication, forms the polynomial ring.

- The zero polynomial is the additive identity
- The constant polynomial 1 is the multiplicative identity.

Example
If $K = \mathbb{R}$, then

$$P(X) = X^2 + 1, \quad Q(X) = X - 3$$

are in $\mathbb{R}[X]$.

$$P + Q = X^2 + X - 2, \quad P \cdot Q = X^3 - 3X^2 + X - 3$$

$\mathbb{R}[X]$ equipped with addition and multiplication is a polynomial ring.

5.3 Arithmetic of Polynomials

5.3.1 Operations on Polynomials

Let K be a commutative ring with identity and let $P, Q \in K[X]$. Addition and multiplication are defined in the previous section we give other operations in this section.

Equality of Polynomials

Let

$$P(X) = a_n X^n + a_{n-1} X^{n-1} + \cdots + a_1 X + a_0$$

and

$$Q(X) = b_n X^n + b_{n-1} X^{n-1} + \cdots + b_1 X + b_0.$$

Then

$$P = Q \iff \forall i \in \{0, \dots, n\},\ a_i = b_i.$$

Scalar Multiplication

For every $\lambda \in K$ and every polynomial

$$P(X) = \sum_{i=0}^{n} a_i X^i,$$

we have

$$(\lambda P)(X) = \sum_{i=0}^{n} (\lambda a_i) X^i.$$

Example

Let

$$P(X) = \alpha X^3 + \beta X^2 + \varepsilon X + \delta \quad \text{and} \quad Q(X) = \alpha X^3 + \gamma X + \alpha.$$

Then

$$\frac{1}{2}P(X) + \frac{1}{2}Q(X) = \alpha X^3 + \frac{1}{2}\beta X^2 + \frac{1}{2}(\varepsilon + \gamma)X + \frac{1}{2}(\delta + \alpha).$$

Properties

- $\deg(P + Q) \leq \max(\deg(P), \deg(Q))$.
- $\deg(P \cdot Q) = \deg(P) + \deg(Q)$.

5.3.2 Divisibility

Definition

Let P and Q be two polynomials with coefficients in K (K is a commutative unital ring), if there is a polynomial $R \in K[X]$ that satisfies:

$$P = Q \cdot R$$

We say that P is divisible by Q or that Q divides P or also P is a multiple of Q and we write $Q \mid P$.

.

Example

Let $P(X) = X^3 - 1$ and $Q(X) = X - 1$, we have

$$X^3 - 1 = (X - 1)(X^2 + X + 1)$$

P is divisible by Q.

5.3.3 Euclidean Division

Theorem 5.3.1. *Let K be a field. Let $P(X) \in K[X]$ be any polynomial and let $S(X) \in K[X]$ be a nonzero polynomial whose leading coefficient is invertible in K. Then there exist unique polynomials $Q(X), R(X) \in K[X]$ such that*

$$P(X) = S(X)Q(X) + R(X), \qquad \deg R < \deg S.$$

The polynomial $Q(X)$ is called the quotient*, and the polynomial $R(X)$ is called the* remainder *of the Euclidean division of $P(X)$ by $S(X)$.*

Example

We divide

$$P(X) = 2X^3 - 3X^2 - 5X + 6 \quad \text{by} \quad S(X) = X - 2$$

in $\mathbb{R}[X]$.

$$\begin{array}{r|l}
2X^3 - 3X^2 - 5X + 6 & X - 2 \\
\hline
2X^2 + X - 3 & \\
\underline{2X^3 - 4X^2} & \\
X^2 - 5X + 6 & \\
\underline{X^2 - 2X} & \\
-3X + 6 & \\
\underline{-3X + 6} & \\
0 &
\end{array}$$

$$2X^3 - 3X^2 - 5X + 6 = (2X^2 + X - 3)(X - 2) + 0.$$

Thus,

$$Q(X) = 2X^2 + X - 3 \quad \text{and} \quad R(X) = 0.$$

5.3.4 GCD and LCM (Pgcd et ppcm)

Definition

Let K be a field and let $P, S \in K[X]$, non-zero polynomial. A greatest common divisor (gcd) is a polynomial $D(X) \in K[X]$ that divides both $P(X)$ and $S(X)$ and for every polynomial $C(X) \in K[X]$, if $C(X)$ divides both $P(X)$ and $S(X)$, then $C(X)$ divides $D(X)$.
When $D(X)$ is chosen with the coefficient of its highest-degree term is exactly 1 , it is unique and is denoted by

$$gcd(P, Q).$$

Euclidean Algorithm

Let A and B be polynomials, $B \neq 0$.

We calculate the successive Euclidean divisions:

$$\begin{aligned}
A &= BQ_1 + R_1 & \deg R_1 &< \deg B \\
B &= R_1Q_2 + R_2 & \deg R_2 &< \deg R_1 \\
R_1 &= R_2Q_3 + R_3 & \deg R_3 &< \deg R_2 \\
\vdots \quad & \quad \vdots & \vdots \\
R_{k-2} &= R_{k-1}Q_k + R_k & \deg R_k &< \deg R_{k-1} \\
R_{k-1} &= R_kQ_{k+1}
\end{aligned}$$

The degree of the remainder decreases with each division. The algorithm stops when the remainder is zero.

The gcd (greatest common divisor) is the last non-zero remainder R_k (made monic).

Definition

Let K be a field and let $P, S \in K[X]$, non-zero polynomial. A least common multiple (LCM) is a polynomial $D(X) \in K[X]$ that is divisible by both $P(X)$ and $S(X)$ and for every polynomial $C(X) \in K[X]$, if $C(X)$ is divisible by both $P(X)$ and $S(X)$, then $C(X)$ is divisible by $D(X)$.
When $D(X)$ is chosen with the coefficient of its highest-degree term is exactly 1 , it is unique and is denoted by

$$lcm(P, Q).$$

Example
Let $P(X) = X^2 - 1, Q(X) = X^2 - 2X + 1$, then

$$\gcd(P, Q) = X - 1, \quad \operatorname{lcm}(P, Q) = X^3 - 2X^2 + X$$

5.3.5 Coprime Polynomials

Definition

Let P and Q two polynomials in $K[X]$ with K is a field. P and Q are coprime if $\gcd(P, Q) = 1$.

5.3.6 Factorization into a product of irreducible factors

Definition

Let K be a field. Polynomials in $K[X]$ that are only divisible by constants (elements of K) and themselves (or their associates, aP with $a \in K^*$) are called irreducible polynomials.

Theorem 5.3.2. *Let K be a field. Every nonconstant polynomial $P(X) \in K[X]$ can be written in the form*

$$P(X) = c\,P_1(X)P_2(X)\cdots P_n(X),$$

where $c \in K^*$ and $P_1(X), \ldots, P_n(X) \in K[X]$ are irreducible polynomials.

Example
$X^2 - 5X + 6 = (X - 2)(X - 3)$ is a factorization into irreducibles over $\mathbb{Q}$.

5.4 Roots of Polynomials

5.4.1 Definition of Roots

Definition

Let $P \in K[X]$ and $a \in K$. We say that a is a root of P if and only if $P(a) = 0$.

Theorem 5.4.1. *Let $P \in K[X]$ and $a \in K$. a is a root of P if and only if P is divisible by $X - a$ in $K[X]$, i.e.:*

$$P(a) = 0 \iff (X - a) \mid P.$$

5.4.2 Roots and Degree

Theorem 5.4.2. *Let $P \in K[X]$ be a polynomial of degree $d \geq 0$. Then P has at most d roots in K.*

In particular, every non-zero polynomial has a finite number of roots.

5.4.3 Multiplicity of Roots

Definition

Multiplicity m is the largest power such that $(X - \alpha)^m$ divides the polynomial P.

Example

$$P(X) = X^2 - 5X + 6 = (X - 2)(X - 3)$$

Roots: 2 and 3, each of multiplicity 1.

$$Q(X) = (X - 1)^3(X + 2)$$

Roots: 1 (multiplicity 3), -2 (multiplicity 1)

Test Your Chapter Mastery

1. Compute $P + Q$ and $P \cdot Q$ for $P(X) = X^2 + 3X + 2$, $Q(X) = X - 1$.
2. Find the remainder and quotient of $X^3 + 2X^2 - 1$ divided by $X + 1$.
3. Factor $X^2 - 7X + 12$ over $\mathbb{Q}$ and find its roots.
4. Determine the multiplicity of the roots of $P(X) = (X - 2)^2(X + 1)$.

6 Solution of exercises

Solution exercise 1
Let P, Q and R be propositions

1. If P is false , Q is true and R is false .

   (a) For the proposition $Q \wedge (P \vee R)$: Since P is false and R is false, the disjunction $P \vee R$ is false. We have Q is true but $P \vee R$ is false hence the conjunction $Q \wedge (P \vee R)$ is false.

   (b) For the proposition $P \vee (Q \wedge R)$: Since Q is true but R is false, the conjunction $Q \wedge R$ is false. Then the disjunction $P \vee (Q \wedge R)$ is false or false, which is false.

   (c) For the proposition $(P \vee Q) \wedge (Q \vee R)$: Since P is false but Q is true, the disjunction $P \vee Q$ is true. Since Q is true, the disjunction $Q \vee R$ is also true. Therefore, the conjunction $(P \vee Q) \wedge (Q \vee R)$ is true and true, which is true. Therefore, proposition (3) is **true**.

   (d) For the proposition $\overline{(P \wedge Q \wedge R)}$: The conjunction $P \wedge Q \wedge R$ is false, because both P and R are false. The negation of this false proposition is true. Therefore, proposition (4) is **true**.

   Therefore, the propositions that are true are (3) and (4).

2. We evaluate each formula using only the truth values that are given.

   (a) For the proposition $Q \Rightarrow (P \Rightarrow R)$, assuming that Q is false:
   An implication with a false antecedent is always true. Therefore, proposition (1) is **true**.

   (b) For the proposition $P \wedge (P \vee Q)$, assuming that Q is true:
   If Q is true, then the disjunction $P \vee Q$ is also true. Thus, the whole expression simplifies to $P \wedge$ true. Therefore, the truth of proposition (2) depends entirely on P. It is true exactly when P is true, and false otherwise.

   (c) For the proposition $P \vee (Q \Rightarrow R)$, assuming that Q is false:
   If Q is false, then the implication $Q \Rightarrow R$ is automatically true. So the disjunction $P \vee (Q \Rightarrow R)$ is $P \vee$ true, which is always true. Therefore, proposition (3) is **true**.

(d) For the proposition $(Q \wedge P) \wedge (\overline{P} \vee R \vee \overline{Q})$, assuming that P is false:
If P is false, then the conjunction $Q \wedge P$ is false. Hence the first part of $(Q \wedge P) \wedge (\overline{P} \vee R \vee \overline{Q})$ is false and we know that a conjunction with a false part is always false. Therefore, proposition (4) is **false**.

3. i. **Truth table for** $P_1 : (P \vee Q) \Rightarrow Q$

P	Q	$P \vee Q$	$(P \vee Q) \Rightarrow Q$
F	F	F	T
F	T	T	T
T	F	T	F
T	T	T	T

Conclusion: P_1 is **not** a tautology (example: $P = T, Q = F$ gives F).

Truth table for $P_2 : (P \vee \overline{Q}) \vee (\overline{P} \wedge Q)$

P	Q	$P \vee \overline{Q}$	$\overline{P} \wedge Q$	P_2
F	F	T	F	T
F	T	F	T	T
T	F	T	F	T
T	T	T	F	T

Conclusion: P_2 is a **tautology** (every row gives T).

Truth table for $P_3 : (P \Rightarrow Q) \vee (Q \Rightarrow P)$

P	Q	$P \Rightarrow Q$	$Q \Rightarrow P$	P_3
F	F	T	T	T
F	T	T	F	T
T	F	F	T	T
T	T	T	T	T

Conclusion: P_3 is a **tautology** (every row gives T).

Truth table for $P_4 : P \wedge (Q \underline{\vee} R) \Leftrightarrow (P \wedge Q) \underline{\vee} (P \wedge R)$

P	Q	R	$Q \underline{\vee} R$	$P \wedge (Q \underline{\vee} R)$	$P \wedge Q$	$P \wedge R$	$(P \wedge Q) \underline{\vee} (P \wedge R)$	P_4
F	F	F	F	F	F	F	F	T
F	F	T	T	F	F	F	F	T
F	T	F	T	F	F	F	F	T
F	T	T	F	F	F	F	F	T
T	F	F	F	F	F	F	F	T
T	F	T	T	T	F	T	T	T
T	T	F	T	T	T	F	T	T
T	T	T	F	F	T	T	F	T

Conclusion: P_4 (the biconditional) is a **tautology** (every row gives T).

ii. Compute simplified negations.

$$\overline{P_1} \Leftrightarrow \overline{(P \vee Q) \Rightarrow Q} \Leftrightarrow \overline{\overline{(P \vee Q)} \vee Q} \Leftrightarrow (P \vee Q) \wedge \overline{Q} \Leftrightarrow P \wedge \overline{Q}.$$

$$\overline{P_2} \Leftrightarrow \overline{(P \vee \overline{Q}) \vee (\overline{P} \wedge Q)} \Leftrightarrow \overline{P \vee \overline{Q}} \wedge \overline{\overline{P} \wedge Q} \Leftrightarrow (\overline{P} \wedge Q) \wedge (P \vee \overline{Q})$$

$$\overline{P_3} \Leftrightarrow \overline{(P \Rightarrow Q) \vee (Q \Rightarrow P)} \Leftrightarrow \overline{(\overline{P} \vee Q)} \wedge \overline{(\overline{Q} \vee P)} \Leftrightarrow (P \wedge \overline{Q}) \wedge (Q \wedge \overline{P})$$

4. i. We prove the biconditional propositions

$$(\overline{P \wedge Q}) \Leftrightarrow \overline{P} \vee \overline{Q}, \qquad (\overline{P \vee Q}) \Leftrightarrow \overline{P} \wedge \overline{Q}, \qquad P \Rightarrow Q \Leftrightarrow \overline{Q} \Rightarrow \overline{P}.$$

De Morgan 1: $\overline{P \wedge Q} \Leftrightarrow \overline{P} \vee \overline{Q}$

P	Q	$P \wedge Q$	$\overline{P \wedge Q}$	$\overline{P} \vee \overline{Q}$
F	F	F	T	T
F	T	F	T	T
T	F	F	T	T
T	T	T	F	F

Both columns coincide in all rows, hence De Morgan law is proved.

De Morgan 2: $\overline{P \vee Q} \Leftrightarrow \overline{P} \wedge \overline{Q}$

P	Q	$P \vee Q$	$\overline{P \vee Q}$	$\overline{P} \wedge \overline{Q}$
F	F	F	T	T
F	T	T	F	F
T	F	T	F	F
T	T	T	F	F

Both columns coincide in all rows; De Morgan law is proved.

Contrapositive: $P \Rightarrow Q \Leftrightarrow \overline{Q} \Rightarrow \overline{P}$

P	Q	$P \Rightarrow Q$	$\overline{Q} \Rightarrow \overline{P}$
F	F	T	T
F	T	T	T
T	F	F	F
T	T	T	T

Both columns coincide in every row. This proves the biconditional proposition.

ii. From the truth tables we see that both sides of each biconditional have identical truth values in all cases. We may therefore deduce that the biconditionals always hold, and consequently each biconditional proposition is a tautology.

5.

Case 1: $P \vee Q$ with $P \vee (Q \Rightarrow R)$.

P	Q	R	$Q \Rightarrow R$	$P \vee Q$	$P \vee (Q \Rightarrow R)$
T	T	T	T	T	T
T	T	F	F	T	T
T	F	T	T	T	T
T	F	F	T	T	T
F	T	T	T	T	T
F	T	F	F	T	F
F	F	T	T	F	T
F	F	F	T	F	T

The two propositions are not equivalent.

Case 2: $P \vee P$ with P.

P	$P \vee P$	P
T	T	T
F	F	F

Conclusion: $P \vee P$ is equivalent to P

Solution exercise 2

Let f be a function defined from $\mathbb{R}$ to $\mathbb{R}$. We have

1. $\forall C \in \mathbb{R}, \exists x \in \mathbb{R} : f(x) \neq C$.
2. $\forall x, y \in \mathbb{R} : (x \leq y \Rightarrow f(x) \geq f(y))$.

3. $\forall x \in \mathbb{R} : (f(-x) = f(x))$.
4. $\exists M \in \mathbb{R}, \exists m \in \mathbb{R}, \forall x \in \mathbb{R} : m \leq f(x) \leq M$.

Solution exercise 3

Let f be a function defined from $I \subset \mathbb{R}$ to $\mathbb{R}$. We have:

1. f is constant.
2. f is periodic.
3. f is strictly increasing.
4. f never has the same value at two distinct points.

Solution exercise 4

1. The negation of each proposition.
   - $\forall x \in \mathbb{R}, \exists y \in \mathbb{R} :\ x + y < 0$.
   - $\exists x \in \mathbb{R}, \forall y \in \mathbb{R} :\ x + y < 0$.
   - $\exists n \in \mathbb{N}, \forall k \in \mathbb{N} :\ n \neq k^2 + 1$.
   - $\exists n \in \mathbb{N}, \forall k \in \mathbb{N} :\ n \neq 2k + 1$.
   - $\forall x \in \mathbb{R}, \exists y \in \mathbb{R} :\ y^2 \leq x$.
   - $\exists k \in \mathbb{Z} :\ (k \leq -2\ \vee\ k \geq 4)$.
2. The truth value of each proposition with justification.
   - "$\exists x \in \mathbb{R}, \forall y \in \mathbb{R} : x + y \geq 0$" is false.
     Because, For an arbitrary $x \in \mathbb{R}$, we can take $y = -x - 1$, and we obtain $x + y = x - x - 1 = -1 < 0$ then, y does not verify the predicate of this proposition(we have just proven that the negation of this proposition is true).
   - "$\forall x \in \mathbb{R}, \exists y \in \mathbb{R} : x + y \geq 0$" is true.
     Indeed, if we consider any $x \in \mathbb{R}$ we can take for example $y = -x + 1$ and we obtain $x + y = x - x + 1 = 1 \geq 0$.
   - "$\forall n \in \mathbb{N}, \exists k \in \mathbb{N} : n = k^2 + 1$" is false.
     Indeed, if we choose $n = 3$, then for every $k \in \mathbb{N}$, we have $3 \neq k^2 + 1$. Thus, the negation of this proposition is true, and therefore the proposition is false.
   - "$\forall n \in \mathbb{N}, \exists k \in \mathbb{N} : n = 2k + 1$" is false.
     This claims that every natural number is odd. But if we take for example $n = 4$, we obtain : $4 = 2 \times 2$ cannot be written as $2k + 1$ with $k \in \mathbb{N}$. Thus, the negation of this proposition is true, and therefore the proposition is false.
   - "$\exists x \in \mathbb{R}, \forall y \in \mathbb{R} : y^2 > x$" is true.
     We choose any $x < 0$ (for example $x = -1$). For every real y we have $y^2 \geq 0 > -1$, hence $y^2 > x$. Therefore such an x exists.
   - "$\forall k \in \mathbb{Z} : -2 < k < 4$" is false.
     The proposition claims every integer k satisfies $-2 < k < 4$. But there are integers outside this interval. It is enough to give a value to k which does not verify the predicate (outside this interval). For example we take $k = 4$ then we have k verify the negation of the predicate: $-2 < k < 4$ (we have just proven that the negation of this proposition is true).

Solution exercise 5

We prove the following proposition using induction proof

$$\forall n \in \mathbb{N},\ \forall x > 0 : (1+x)^n \geq 1 + nx.$$

For $n \in \mathbb{N}$, we denote by $P(n)$ the statement:

$$\forall x > 0 : (1+x)^n \geq 1 + nx.$$

We want to prove that $\forall n \in \mathbb{N} : P(n)$ is true.

The Base Case: Let us check $P(0)$. For $n = 0$, for all $x > 0$ we have

$$(1+x)^0 = 1 \geq 1,$$

Hence, for all $x > 0$: $P(0)$ is true.

The Inductive Step: Let $n \in \mathbb{N}$. We suppose that $P(n)$ is true, that is

$$\forall x > 0 :\ (1+x)^n \geq 1 + nx.$$

and we prove that $P(n+1)$ holds .i.e.

$$\forall x > 0 :\ (1+x)^{n+1} \geq 1 + (n+1)x.$$

For all $x > 0$, we have

$$(1+x)^{n+1} = (1+x)(1+x)^n.$$

By the induction hypothesis,

$$(1+x)^n \geq 1 + nx.$$

Therefore,

$$(1+x)^{n+1} \geq (1+nx)(1+x).$$

Now, expanding gives

$$(1+nx)(1+x) = 1 + (n+1)x + nx^2 \geq 1 + (n+1)x,$$

since $nx^2 \geq 0$ for all $n \in \mathbb{N}$ and $x > 0$.

By transitivity,

$$(1+x)^{n+1} \geq 1 + (n+1)x.$$

Thus, $P(n+1)$ is true.

The Conclusion: From the two previous steps and by the principle of mathematical induction, we conclude that

$$\forall n \in \mathbb{N},\ \forall x > 0 :\ (1+x)^n \geq 1 + nx.$$

Solution exercise 6

1. We prove that $\sqrt{2}$ is irrational i.e. $\sqrt{2} \notin \mathbb{Q}$ using the proof by contradiction (reasoning by absurdity).
   We assume that $\sqrt{2}$ is rational i.e. $\sqrt{2} \in \mathbb{Q}$ (the negation of the proposition).
   Then, let
   $$\sqrt{2} = \frac{p}{q}, \tag{6.1}$$
   where $p, q \in \mathbb{Z}$, with $q \neq 0$.
   Also we assume that p and q are relatively prime i.e $\gcd(p, q) = 1$.
   We square both sides of (6.1), we get
   $$2 = \frac{p^2}{q^2},$$
   then,
   $$p^2 = 2q^2, \tag{6.2}$$
   therefore, p^2 is even, then
   $$p \text{ is even, i.e. } p = 2m \ /m \in \mathbb{Z},$$
   Replacing p by $2m$ in (6.2), we obtain
   $$4m^2 = 2q^2,$$
   by simplification, we get
   $$2m^2 = q^2,$$
   therefore, q^2 is even, then q is even.
   We conclude that both p and q are even then, have 2 as a common factor. But we assumed that p and q are relatively prime. This is a contradiction. Thus $\sqrt{2}$ is irrational i.e. $\sqrt{2} \notin \mathbb{Q}$.

2. We prove that $\sqrt{4 - 2\sqrt{3}} - \sqrt{4 + 2\sqrt{3}}$ is a rational number. We pose
   $$x = \sqrt{4 - 2\sqrt{3}} - \sqrt{4 + 2\sqrt{3}},$$
   then
   $$x^2 = (4 - 2\sqrt{3}) + (4 + 2\sqrt{3}) - 2\sqrt{(4 - 2\sqrt{3})(4 + 2\sqrt{3})},$$
   hence
   $$x^2 = 8 - 2\sqrt{4} = 8 - 4 = 4.$$
   Furtheremore, $x \leq 0$, then
   $$x = -2.$$
   Therefore, $\sqrt{4 - 2\sqrt{3}} - \sqrt{4 + 2\sqrt{3}}$ is a rational number.

Solution exercise 7

We want to prove the following property, for $n \in \mathbb{N}^*$:

$$\text{If } (n^2 - 1) \text{ is not divisible by 8, then } n \text{ is even.}$$

1. If n is odd, then $n^2 - 1$ is divisible by 8.

2. Let us take n an odd integer. Then n can be written as
$$n = 2l + 1 \quad \text{with } l \in \mathbb{N}.$$
If l is even, $l = 2k$ with $k \in \mathbb{N}$ and so $n = 4k + 1$. If l is odd, $l = 2k + 1$ with $k \in \mathbb{N}$ and so $n = 4k + 3$. In both cases, we can write
$$n = 4k + r \quad \text{with } k \in \mathbb{N},\ r \in \{1, 3\}.$$
Squaring gives:
$$n^2 - 1 = (4k + r)^2 - 1 = 16k^2 + 8kr + r^2 - 1 = 8(2k^2 + kr) + (r^2 - 1).$$
If $r = 1$, then $r^2 - 1 = 0$ and thus $n^2 - 1$ is divisible by 8. If $r = 3$, then $r^2 - 1 = 8$ and thus $n^2 - 1$ is also divisible by 8.

3. Yes. Since the contrapositive of an implication is logically equivalent to the proposition itself, proving the contrapositive proves the original proposition.

Solution exercise 8

Let P be the proposition:
$$\exists a \in]-\infty, 0[, \forall x \in [-1, +\infty[: \ \Big(x > a \implies \Big[(x^2 > \tfrac{a^2}{4}) \ \text{ or } \ (x \leq 0)\Big]\Big).$$
To prove P is false we give a counter example i.e. The example that we give, it prove that the negation of P is true. The negation of P is
$$\forall a \in]-\infty, 0[, \ \exists x \in [-1, +\infty[, \ \Big(x > a \ \text{ and } \ (x^2 \leq \tfrac{a^2}{4}) \ \text{ and } \ (x > 0)\Big).$$
Let $a < 0$ and we take
$$x = -\tfrac{a}{2}.$$
Clearly, $x > 0$ and, in particular, $x > a$. Moreover, we have
$$x^2 = \frac{a^2}{4} \ \leq \ \frac{a^2}{4}.$$
Therefore, This example prove that the negation of P is true, and it it is a counter example for the proposition P, which means that P is false.

Solution exercise 9

We prove that: $\forall n \in \mathbb{N}: \ n(n + 1)$ is divisible by 2 using case by case method.

Case 1: for n an even number : $n = 2k$ with $k \in \mathbb{N}$ so $n(n + 1) = 2k(2k + 1) = 2\,(2k^2 + k)$, we note $k' = (2k^2 + k)$ with $k' \in \mathbb{N}$ then $n(n + 1)$ is divisible by 2.

Case 2: for n an odd number : $n = 2k + 1$ with $k \in \mathbb{N}$ so $n(n + 1) = (2k + 1)(2k + 2) = 2(2k + 1)(k + 1)$, we note $k'' = (2k + 1)(k + 1)$ with $k'' \in \mathbb{N}$ then $n(n + 1)$ is divisible by 2.

Therefore, in all cases the predicate is true. Solution exercise 10

Let $A = \{\{-2\}, \{3\}, \{4\}\}$ and $B = \{\{3\}, \{-2, 3, 4\}\}$.

(a) We determine the following sets:

$$A \cap B = \{\{3\}\},\ A \cup B = \{\{-2\}, \{3\}, \{4\}, \{-2, 3, 4\}\},\ A\Delta B = \{\{-2\}, \{4\}, \{-2, 3, 4\}\}$$

$B - A = \{\{-2, 3, 4\}\}$,
$\mathcal{P}(A) = \Big\{ \ \emptyset,\ A,\ \{\{-2\}\}, \{\{3\}\}, \{\{4\}\}, \{\{-2\}, \{3\}\}, \{\{-2\}, \{4\}\}, \{\{3\}, \{4\}\} \ \Big\}$,
$A \times B = \{(\{-2\}, \{3\}), (\{-2\}, \{-2, 3, 4\}), (\{3\}, \{3\}), (\{3\}, \{-2, 3, 4\}),$
$(\{4\}, \{3\}), (\{4\}, \{-2, 3, 4\})\}$.

(b) We replace the points... by one of the following symbols : $\in$, $\subset$, $\notin$.
$\{3\} \in A \cap B$, $\{\{-2\}\} \subset A \cup B$, $\{-2,3,4\} \in A\Delta B$, $\{-2\} \notin B - A$, $\{A\} \subset \mathcal{P}(A)$, $(\{-2\},\{5\}) \notin A \times B$.

(c) Let the universal set be $E = A \cup (B \cup \{\{5\}\})$.

(a) $\complement_E A = \{\{-2,3,4\},\{5\}\}$..

(b) A partition for the set E

$$\mathcal{G} = \{\, A,\ \{\{-2,3,4\},\{5\}\}\,\} = \Big\{\{\{-2\},\{3\},\{4\}\},\ \{\{-2,3,4\},\{5\}\}\Big\}.$$

all the conditions are verified for $\mathcal{G}$.

Solution exercise 11

1. • We have
$$E = \{x \in \mathbb{Z} \mid x^2 - 5x + 6 = 0 \text{ or } -2 < x \leq 2\}.$$
This means that an integer x belongs to E if it satisfies $x^2 - 5x + 6 = 0$ or if it satisfies the inequality $-2 < x \leq 2$. Hence, E is the union of the two corresponding sets:
$$E = \{x \in \mathbb{Z} \mid x^2 - 5x + 6 = 0\} \cup \{x \in \mathbb{Z} \mid -2 < x \leq 2\}.$$
The first set gives $\{2,3\}$, and the second gives $\{-1,0,1,2\}$. Taking the union, we get
$$E = \{-1,0,1,2,3\}.$$

• We consider another finite set F, we have:
$$|\mathcal{P}(F)| = 32 = 2^5$$
then,
$$|F| = 5$$
also we have
$$|E \cup F| = 9 = |E| + |F| - |E \cap F| = 5 + 5 - |E \cap F|$$
Therefore,
$$|E \cap F| = 1$$

2. We prove that $A = C$, i.e.
$$\forall (x,y) \in \mathbb{R}^2 :\ [(x,y) \in A \iff (x,y) \in C].$$
This requires both inclusions $C \subseteq A$ and $A \subseteq C$.

(1) $C \subseteq A$.
$$\forall (x,y) \in \mathbb{R}^2 :\ [(x,y) \in C \Rightarrow (x,y) \in A],$$
equivalently,
$$\forall (x,y) \in \mathbb{R}^2 :\ [(\exists t \in \mathbb{R},\ x = t+1 \wedge y = 4t+3) \Rightarrow (4x - y = 1)].$$
Let $(x,y) \in C$. Then there exists $t \in \mathbb{R}$ such that $x = t+1$ and $y = 4t+3$. Thus
$$4x - y = 4(t+1) - (4t+3) = 4t + 4 - 4t - 3 = 1,$$

so $(x, y) \in A$.

(2) $A \subseteq C$.

$$\forall (x, y) \in \mathbb{R}^2,\ [(x, y) \in A \Rightarrow (x, y) \in C],$$

equivalently,

$$\forall (x, y) \in \mathbb{R}^2 :\ [(4x - y = 1) \Rightarrow (\exists t \in \mathbb{R},\ x = t + 1 \wedge y = 4t + 3)].$$

Let $(x, y) \in A$. Then $4x - y = 1$, i.e. $y = 4x - 1$. For an arbitrary $x \in A$, we necessarily take $t = x - 1$ and we obtain $x = t + 1$ and

$$y = 4x - 1 = 4(x - 1) + 3 = 4t + 3,$$

Hence, $(x, y) = (t + 1, 4t + 3) \in C$.

Since $C \subseteq A$ and $A \subseteq C$, we obtain

$$\boxed{A = C.}$$

3. Let

$$D = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 1\}.$$

We use the proof by contradiction. We suppose that there exist $A, B \subset \mathbb{R}$ such that $D = A \times B$. Then the point $(1, 0) \in D$, and therefore $1 \in A$. Similarly, $(0, 1) \in D$, and therefore $1 \in B$. We can then deduce that $(1, 1) \in A \times B = D$, but $(1, 1) \notin D$ (because $1^2 + 1^2 = 2 > 1$) which is a contradiction. Hence, D is not the Cartesian product of two subsets of $\mathbb{R}$.

Solution exercise 12

1. Let $A, B, C \subseteq E$.

   (a) We simplify the following sets:

   $$(\complement_E(A \cup B)) \cap (\complement_E(C \cup \complement_E A))$$

   Using De Morgan law:

   $$\complement_E(A \cup B) = \complement_E A \cap \complement_E B, \qquad \complement_E(C \cup \complement_E A) = \complement_E C \cap A.$$

   Hence

   $$(\complement_E A \cap \complement_E B) \cap (\complement_E C \cap A) = (\complement_E A \cap A) \cap (\complement_E B \cap \complement_E C) = \varnothing.$$

   $$(\complement_E(A \cap \complement_E B)) \cup (\complement_E(C \cap \complement_E A))$$

   By De Morgan law:

   $$\complement_E(A \cap \complement_E B) = \complement_E A \cup B, \qquad \complement_E(C \cap \complement_E A) = \complement_E C \cup A.$$

   Thus

   $$(\complement_E A \cup B) \cup (\complement_E C \cup A) = (\complement_E A \cup A) \cup (B \cup \complement_E C) = E.$$

   (b) We prove the properties.

- $A \subseteq B \Longleftrightarrow \complement_E B \subseteq \complement_E A$.

$$\begin{aligned} A \subseteq B &\Longleftrightarrow \forall x \in E\,(x \in A \Rightarrow x \in B) \\ &\Longleftrightarrow \forall x \in E\,(\overline{(x \in B)} \Rightarrow \overline{(x \in A)}) \\ &\Longleftrightarrow \forall x \in E\,(x \notin B \Rightarrow x \notin A) \\ &\Longleftrightarrow \forall x \in E\,\Big(x \in \complement_E B \Rightarrow x \in \complement_E A\Big) \\ &\Longleftrightarrow \complement_E B \subseteq \complement_E A. \end{aligned}$$

- $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

Let $x \in E$, We have

$$\begin{aligned} x \in A \cap (B \cup C) &\Longleftrightarrow x \in A \ \wedge\ (x \in B \vee x \in C) \\ &\Longleftrightarrow (x \in A \wedge x \in B) \ \vee\ (x \in A \wedge x \in C) \\ &\Longleftrightarrow x \in (A \cap B) \ \vee\ x \in (A \cap C) \\ &\Longleftrightarrow x \in (A \cap B) \cup (A \cap C). \end{aligned}$$

Therefore,

$$\forall x \in E:\ (x \in A \cap (B \cup C) \Longleftrightarrow x \in (A \cap B) \cup (A \cap C)).$$

Thus, $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

- If $A \cap B = A \cup B$ then $A = B$.

Let $x \in E$ and we suppose$x \in A$

$$\begin{aligned} x \in A &\Rightarrow x \in A \cup B \\ &\Rightarrow x \in A \cap B \quad (\text{since } A \cup B = A \cap B) \\ &\Rightarrow x \in B. \end{aligned}$$

Thus, $\forall x \in E:\ (x \in A \Rightarrow x \in B)$.

Hence $A \subseteq B$.

Now we suppose $x \in B$, we get

$$\begin{aligned} x \in B &\Rightarrow x \in A \cup B \\ &\Rightarrow x \in A \cap B \\ &\Rightarrow x \in A. \end{aligned}$$

Thus, $\forall x \in E:\ (x \in B \Rightarrow x \in A)$.

Hence $B \subseteq A$.

Therefore, we obtain the double inclusion which gives the equality $A = B$.

- $A - B = A \cap \complement_E B$.

Let $x \in E$ we have

$$\begin{aligned} x \in A - B &\Longleftrightarrow x \in A \ \wedge\ x \notin B \\ &\Longleftrightarrow x \in A \ \wedge\ x \in \complement_E B \\ &\Longleftrightarrow x \in A \cap \complement_E B. \end{aligned}$$

Hence,

$$\forall x \in E:\ x \in A - B \Leftrightarrow x \in A \cap \complement_E B.$$

Which gives the equality.

- We want to prove that the intersection is distributive under the difference, that is:
$$A \cap (B - C) = (A \cap B) - (A \cap C).$$
We have:
$$B - C = B \cap \complement_E C.$$
Then,
$$A \cap (B - C) = A \cap (B \cap \complement_E C) = (A \cap B) \cap \complement_E C.$$
On the other hand,
$$(A \cap B) - (A \cap C) = (A \cap B) \cap \complement_E (A \cap C).$$
We have:
$$\complement_E (A \cap C) = \complement_E A \cup \complement_E C.$$
Hence,
$$(A \cap B) - (A \cap C) = (A \cap B) \cap (\complement_E A \cup \complement_E C) = [(A \cap B) \cap \complement_E A] \cup [(A \cap B) \cap \complement_E C].$$
The first term is empty because $(A \cap B) \cap \complement_E A = \varnothing$, so
$$(A \cap B) - (A \cap C) = (A \cap B) \cap \complement_E C.$$
Therefore,
$$A \cap (B - C) = (A \cap B) - (A \cap C).$$
- $A \cap (B \triangle C) = (A \cap B) \triangle (A \cap C)$.
Recall that $B \triangle C = (B - C) \cup (C - B)$. Then using the distributivity of $\cap$ under $\cup$, we obtain
$$A \cap (B \triangle C) = (A \cap (B - C)) \cup (A \cap (C - B))$$
$$= ((A \cap B) - (A \cap C)) \cup ((A \cap C) - (A \cap B)) = (A \cap B) \triangle (A \cap C).$$

(c) .

i. We have:
$$((A \subseteq A \cup B) \wedge (B \subseteq A \cup B)) \implies ((\mathcal{P}(A) \subseteq \mathcal{P}(A \cup B)) \wedge (\mathcal{P}(B) \subseteq \mathcal{P}(A \cup B)))$$
$$\implies (\mathcal{P}(A) \cup \mathcal{P}(B)) \subseteq \mathcal{P}(A \cup B).$$
The equality is not generally true. We give an example:
Let $A = \{a\}$ and $B = \{b\}$ be two sets. Therefore, $A \cup B = \{a, b\}$ and $\mathcal{P}(A) = \{\emptyset, \{a\}\}$, $\mathcal{P}(B) = \{\emptyset, \{b\}\}$ and $\mathcal{P}(A \cup B) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$. We observe that, $\mathcal{P}(A) \cup \mathcal{P}(B) = \{\emptyset, \{a\}\} \cup \{\emptyset, \{b\}\} = \{\emptyset, \{a\}, \{b\}\} \subset \{\emptyset, \{a\}, \{b\}, \{a, b\}\} = \mathcal{P}(A \cup B)$.

ii. On one hand, we have:
$$((A \cap B \subseteq A) \wedge (A \cap B \subseteq B)) \implies (\mathcal{P}(A \cap B) \subseteq \mathcal{P}(A)) \wedge (\mathcal{P}(A \cap B) \subseteq \mathcal{P}(B))$$
$$\implies \mathcal{P}(A \cap B) \subseteq \mathcal{P}(A) \cap \mathcal{P}(B).$$
and on the other hand, we have: let $C \in \mathcal{P}(A) \cap \mathcal{P}(B)$ then $C \in \mathcal{P}(A)$ and $C \in \mathcal{P}(B)$ consequently $C \subseteq A$ and $C \subseteq B$ we deduce that $C \subseteq A \cap B$ therefore $C \in \mathcal{P}(A \cap B)$. Hence, $\forall C : (C \in \mathcal{P}(A) \cap \mathcal{P}(B) \implies C \in \mathcal{P}(A \cap B))$ Therefore, $\mathcal{P}(A) \cap \mathcal{P}(B) \subseteq \mathcal{P}(A \cap B)$.
Finally, we conclude the equality $\mathcal{P}(A) \cap \mathcal{P}(B) = \mathcal{P}(A \cap B)$.

2. Let E be a finite universal set, and $A, B, C, D \subseteq E$.

   We prove that:

$$|\complement_{E\times E}((A\times B)\triangle(C\times D))| = |E|^2 - |A\times B| - |C\times D| + 2|(A\times B)\cap(C\times D)|.$$

   We pose $X = A\times B$ and $Y = C\times D$. Then:

$$|\complement_{E\times E}(X\triangle Y)| = |E^2| - |X\triangle Y| = |E|^2 - |X\triangle Y|.$$

   Furtheremore,

$$\begin{aligned}
|X\triangle Y| &= \Big|(X-Y)\cup(Y-X)\Big| \\
&= |X-Y| + |Y-X| - |(X-Y)\cap(Y-X)\Big| \quad (|(X-Y)\cap(Y-X)\Big| = 0 \text{ because are disjoint}) \\
&= \Big(|X| - |X\cap Y|\Big) + \Big(|Y| - |X\cap Y|\Big) \\
&= |X| + |Y| - 2|X\cap Y|.
\end{aligned}$$

   Therefore:

$$|\complement_{E\times E}(X\triangle Y)| = |E|^2 - |X| - |Y| + 2|X\cap Y|.$$

   Since $|A\times B| = |A|\cdot|B|$ and $|C\times D| = |C|\cdot|D|$, we get:

$$|\complement_{E\times E}((A\times B)\triangle(C\times D))| = |E|^2 - |A||B| - |C||D| + 2|(A\times B)\cap(C\times D)|.$$

Solution exercise 13

1. Let $f : E \longrightarrow F$ be a function. For $A, B \subseteq E$ and $C, D \subseteq F$. We prove the following properties:

   - $f(A\cup B) = f(A)\cup f(B)$.

     We procced by double inclusion
     (a) $f(A)\cup f(B) \subseteq f(A\cup B)$

     We know that:

$$(A\subseteq A\cup B)\wedge(B\subseteq A\cup B) \Rightarrow f(A)\subseteq f(A\cup B) \text{ and } f(B)\subseteq f(A\cup B)$$

     We used the property ($G\subseteq H \Rightarrow f(G)\subseteq f(H)$. that is proved in the course session).
     Therefore,

$$f(A)\cup f(B) \subseteq f(A\cup B)$$

     (b) $f(A\cup B)\subseteq f(A)\cup f(B)$
     Let $y\in F$, we suppose that $y\in f(A\cup B)$, then $\exists x\in A\cup B:\ y = f(x)$.
     Since $x\in A\cup B$, we have $x\in A$ or $x\in B$. Hence,

$$(\exists x\in A \text{ such that } y = f(x)) \vee (\exists x\in B \text{ such that } y = f(x))$$

     which implies

$$y\in f(A) \text{ or } y\in f(B)$$

and consequently,

$$y \in f(A) \cup f(B)$$

Thus,

$$f(A \cup B) \subseteq f(A) \cup f(B)$$

From (a) and (b), we conclude that

$$f(A \cup B) = f(A) \cup f(B)$$

- If f is injective then $f(A \cap B) = f(A) \cap f(B)$.
  We know that

  $$A \cap B \subseteq A \quad \text{and} \quad A \cap B \subseteq B,$$

  then

  $$f(A \cap B) \subseteq f(A) \quad \text{and} \quad f(A \cap B) \subseteq f(B).$$

  Hence, $f(A \cap B)$ is contained in both $f(A)$ and $f(B)$. Therefore,

  $$f(A \cap B) \subseteq f(A) \cap f(B),$$

  this inclusion is always hold even if f is non-injective.
  Next, we prove that

  $$f(A) \cap f(B) \subseteq f(A \cap B)$$

  Let $y \in F$ and we suppose that $y \in f(A) \cap f(B)$. Then $y \in f(A)$ and $y \in f(B)$, so there exist $x \in A$ and $x' \in B$ such that

  $$y = f(x) = f(x')$$

  Since f is injective, we deduce that $x = x'$. Therefore, $\exists x \in A \cap B : \ y = f(x)$, and consequently,

  $$y \in f(A \cap B)$$

  Thus,

  $$f(A) \cap f(B) \subseteq f(A \cap B)$$

  Combining both inclusions, we conclude if f is injective then

  $$f(A \cap B) = f(A) \cap f(B).$$

- $f^{-1}(C \cup D) = f^{-1}(C) \cup f^{-1}(D)$.
  Let $x \in E$. We suppose $x \in f^{-1}(C \cup D)$. We have

  $$\begin{aligned} x \in f^{-1}(C \cup D) &\iff f(x) \in C \cup D \\ &\iff f(x) \in C \ \vee \ f(x) \in D \\ &\iff x \in f^{-1}(C) \ \vee \ x \in f^{-1}(D) \\ &\iff x \in f^{-1}(C) \cup f^{-1}(D). \end{aligned}$$

  We proved $\forall x \in E : \ (x \in f^{-1}(C \cup D) \Leftrightarrow x \in f^{-1}(C) \cup f^{-1}(D))$. Therefore, the two sets are equal.

- $f^{-1}(C \cap D) = f^{-1}(C) \cap f^{-1}(D)$.
  Let $x \in E$. We suppose $x \in f^{-1}(C \cap D)$. We have

$$\begin{aligned} x \in f^{-1}(C \cap D) &\iff f(x) \in C \cap D \\ &\iff f(x) \in C \ \wedge \ f(x) \in D \\ &\iff x \in f^{-1}(C) \ \wedge \ x \in f^{-1}(D) \\ &\iff x \in f^{-1}(C) \cap f^{-1}(D). \end{aligned}$$

  We proved $\forall x \in E: \ (x \in f^{-1}(C \cap D) \Leftrightarrow x \in f^{-1}(C) \cap f^{-1}(D))$. Therefore, the two sets are equal.

2. Let

$$f : \mathbb{R} \longrightarrow \mathbb{R}, \qquad f(x) = x^2 + 1.$$

   (a) We determine the direct images $f([-2,1])$, $f([-2,1] \cup [2,+\infty[)$ and $f(\mathbb{R})$.

   (i) $f([-2,1]) = [1,5]$.
   Let $y \in f([-2,1])$, then

$$y \in f([-2,1]) \Leftrightarrow \exists x \in [-2,1] : \ y = f(x) = x^2 + 1$$

   We have $x \in [-2,1]$. We divide the interval where f is monotonic
   If $x \in [-2,0]$ then $-2 \leq x \leq 0$ thus $0 \leq x^2 \leq 4$, hence

$$1 \leq x^2 + 1 \leq 5$$

   therefore,

$$y = x^2 + 1 \in [1,5],$$

   If $x \in [0,1]$, then $0 \leq x \leq 1$ thus $0 \leq x^2 \leq 1$, hence

$$1 \leq x^2 + 1 \leq 2$$

   thus,

$$y = x^2 + 1 \in [1,2],$$

   Therefore, $y \in [1,5]$ or $y \in [1,2]$ then $y \in [1,5] \cup [1,2] = [1,5]$. Hence $f([-2,1]) \subseteq [1,5]$.

   Conversely, we prove $\forall y \in [1,5], \exists x \in [-2,1] : \ y = f(x) = x^2 + 1$
   For any $y \in [1,5]$, we solve $y = x^2 + 1$, i.e. $x = \pm\sqrt{y-1}$. Since $y - 1 \in [0,4]$ we have $\sqrt{y-1} \in [0,2]$. Then we choose

$$x = -\sqrt{y-1}$$

   We have

$$1 \leq y \leq 5 \Rightarrow -2 \leq -\sqrt{y-1} \leq 0 \Rightarrow -\sqrt{y-1} \in [-2,0] \subset [-2,1]$$

   therefore, $x = -\sqrt{y-1}$ belongs to $[-2,1]$ for every $y \in [1,5]$ such that $f\Big(-\sqrt{y-1}\Big) = y$.

   Hence $[1,5] \subseteq f([-2,1])$.
   Combining the two inclusions gives $f([-2,1]) = [1,5]$.

(ii) $f([-2,1]\cup[2,\infty[) = [1,\infty[$.

We have from the first part of the exercise $f([-2,1]\cup[2,\infty[) = f([-2,1])\cup f([2,\infty[)$ $f([-2,1])$ is obtained we still have to find $f([2,\infty[)$.
Let $y \in f([2,\infty[)$, then

$$y \in f([2,\infty[) \Leftrightarrow \exists x \in [2,\infty[:\ y = f(x) = x^2+1$$

We have $x \in [2,\infty[$ then $x^2 \geq 4$, so $y = f(x) = x^2+1 \geq 5$. Thus $f([2,\infty[) \subseteq [5,\infty[$.

Conversely, we prove $\forall y \in [5,\infty[, \exists x \in [2,\infty[:\ y = f(x) = x^2+1$

For any $y \in [5,\infty[$, we solve $y = x^2+1$, i.e. $x = \pm\sqrt{y-1}$. We have

$$x = -\sqrt{y-1} \notin [5,\infty[$$

We have

$$y \geq 5 \Rightarrow \sqrt{y-1} \geq 2 \Rightarrow \sqrt{y-1} \in [2,\infty[$$

Then we choose $x = \sqrt{y-1} \in [2,\infty[$ for every $y \in [1,5]$ such that $f\left(\sqrt{y-1}\right) = y$.
Hence $[5,\infty[\subseteq f([-2,1])$.
Combining the two inclusions gives $f([2,\infty[) = [5,\infty[$.
We conclude $f([-2,1]\cup[2,\infty[) = f([-2,1])\cup f([2,\infty[) = [1,5]\cup[5,\infty[= [1,\infty[$.

(iii) $f(\mathbb{R}) = [1,\infty[$.

Let $y \in f(\mathbb{R})$, then

$$y \in f(\mathbb{R}) \Leftrightarrow \exists x \in \mathbb{R}:\ y = f(x) = x^2+1$$

For $x \in \mathbb{R}$ we have $x^2 \geq 0$, hence $f(x) = x^2+1 \geq 1$, so $f(\mathbb{R}) \subseteq [1,\infty[$.

Conversely, we prove $\forall y \in [1,\infty[, \exists x \in \mathbb{R}:\ y = f(x) = x^2+1$
For any $y \geq 1$, we solve $y = x^2+1$, i.e. $x = \pm\sqrt{y-1}$.
For any $y \geq 1$, we choose $x = \sqrt{y-1}$ (or $x = -\sqrt{y-1}$) because the two solutions are in $\mathbb{R}$ then $x \in \mathbb{R}$ such that $y = f(x) = x^2+1$.
Thus $[1,\infty[\subseteq f(\mathbb{R})$.
Combining the two inclusions gives $f(\mathbb{R}) = [1,\infty[$.

(b) We determine the inverse images $f^{-1}([0,1])$, $f^{-1}(]-\infty,5]\cup[10,\infty[)$ and $f^{-1}([0,1]\cap[-4,0[)$.

(i) $f^{-1}([0,1]) = \{0\}$.

$$f^{-1}([0,1]) = \{x \in \mathbb{R} \mid x^2+1 \in [0,1]\} = \{x \in \mathbb{R} \mid x^2+1 \geq 0 \wedge x^2+1 \leq 1\}$$

$f(x) = x^2+1 \geq 0$ holds for all $x \in \mathbb{R}$. We solve $x^2+1 \leq 1$ but $x^2+1 < 1$ has no solution in $\mathbb{R}$ then the solutions of this inequality are the solutions of the equation $x^2+1 = 1 \Rightarrow x^2 = 0 \Rightarrow x = 0$. Hence $f^{-1}([0,1]) = \{x \in \mathbb{R} \mid x \in \mathbb{R} \wedge x = 0\} = \{0\}$.

(ii) $f^{-1}(]-\infty,5]\cup[10,+\infty[) =]-\infty,-3] \cup [-2,2] \cup [3,+\infty[$.

We have $f^{-1}(]-\infty,5]\cup[10,+\infty[) = f^{-1}(]-\infty,5]) \cup f^{-1}([10,+\infty[)$

To find $f^{-1}(]-\infty,5]) = \{x \in \mathbb{R} \mid f(x) \in]-\infty,5]\}$ we solve $f(x) \leq 5$ then $x^2+1 \leq 5 \iff x^2 \leq 4 \iff x \in [-2,2]$.

Next, To find $f^{-1}([10,+\infty[) = \{x \in \mathbb{R} \mid f(x) \in [10,+\infty[\}$ we solve $f(x) \geq 10$ is $x^2+1 \geq 10 \iff x^2 \geq 9 \iff x \in]-\infty,-3] \cup [3,+\infty[$.

Therefore, $f^{-1}(]-\infty,5] \cup [10,+\infty[) =]-\infty,-3] \cup [-2,2] \cup [3,+\infty[$.

(iii) $f^{-1}([0,1] \cap [-4,0[) = \varnothing$.
The intersection $[0,1] \cap [-4,0[= \varnothing$. The preimage of the empty set is the empty set, so $f^{-1}(\varnothing) = \varnothing$.

(c) Is f injective? Give a restriction of f that is injective (justify).

f is *not* injective on $\mathbb{R}$: indeed $f(1) = 1^2+1 = 2$ and $f(-1) = (-1)^2+1 = 2$, then $f(1) = f(-1)$ and $1 \neq -1$.
A simple injective restriction is $f|_{[0,\infty[} : [0,\infty[\to \mathbb{R}$ given by $f|_{\mathbb{R}_+}(x) = x^2+1$ with domain $[0,\infty[$.
Justification. Let $a,b \in [0,\infty[$ and we suppose $f|_{\mathbb{R}_+}(a) = f|_{\mathbb{R}_+}(b)$ then $a^2+1 = b^2+1$, so $a^2 = b^2$. Since $a,b \geq 0$ this implies $a = |a| = |b| = b$. Thus $a = b$, so $f|_{[0,\infty[}$ is injective.
(We may also restrict to $]-\infty,0]$; f is injective there as well.)

(d) Let $g : \mathbb{R} \to \mathbb{R}$, $g(x) = 3x-1$. Determine $g \circ f$ and $f \circ g$ (explicit expressions). Are they equal?

$$\begin{aligned} g \circ f : \mathbb{R} &\longrightarrow \mathbb{R}, \\ x &\longmapsto y = f(x) = (g \circ f)(x) \end{aligned}$$

For all $x \in \mathbb{R}$,

$$(g \circ f)(x) = g\Big(f(x)\Big) = g(x^2+1) = 3(x^2+1) - 1 = 3x^2+2,$$

and

$$\begin{aligned} f \circ g : \mathbb{R} &\longrightarrow \mathbb{R}, \\ x &\longmapsto y = f(x) = (f \circ g)(x) \end{aligned}$$

For all $x \in \mathbb{R}$,

$$(f \circ g)(x) = f\Big(g(x)\Big) = f(3x-1) = (3x-1)^2+1 = 9x^2-6x+2.$$

These two functions are not equal: For $x = \frac{1}{2}$:

$$(g \circ f)\Big(\tfrac{1}{2}\Big) = 3\Big(\tfrac{1}{4}\Big) + 2 = \tfrac{3}{4} + 2 = \tfrac{11}{4},$$

while

$$(f \circ g)\Big(\tfrac{1}{2}\Big) = 9\Big(\tfrac{1}{4}\Big) - 6\Big(\tfrac{1}{2}\Big) + 2 = \tfrac{9}{4} - 3 + 2 = \tfrac{1}{4},$$

so they differ; therefore $g \circ f \neq f \circ g$.

Solution exercise 14
Let

$$E = \{m, n, p, q, r, s\}, \qquad F = \{1, 2, 3, 4, 5, 6\}.$$

We define a function $f : E \to F$ as follows:

$$f(m) = 2, \qquad f(p) = 5, \qquad f(r) = 1.$$

For the other elements of E, we know that

$$\{(n, 4), (q, 2), (s, 6)\} \subset Graph(f).$$

1. We determine the direct image of the subset $A = \{m, n, q, s\}$.

$$f(A) = \{f(x) \mid x \in A\} = \{f(m), f(n), f(q), f(s)\} = \{2, 4, 6\}$$

2. We determine the inverse image of the subset $B = \{2, 3, 4, 6\}$. (Please correct the set B for students in the series)

$$f^{-1}(B) = \{x \mid f(x) \in B\} = \{x \mid f(x) = 2 \vee f(x) = 3 \vee f(x) = 4 \vee f(x) = 6\} = \{m, q, n, s\}.$$

3. We determine the range of f.

$$\{y \in F \mid \exists x \in E : y = f(x)\} = \{f(x) \mid x \in E\} = \{2, 4, 5, 1, 6\}$$

4. We have the following property that is true for any function f defined from a set E to a set F

$$f \text{ is surjective} \Leftrightarrow f(E) = F.$$

Using this property and from the previous question we have $f(E) = \{f(x) \mid x \in E\} = \{2, 4, 5, 1, 6\} \neq F$, therefore the function f –defined in the exercise– is not surjective.

Solution exercise 15
Let f be a function defined by:

$$f : \mathbb{R} - \left\{\frac{1}{2}\right\} \longrightarrow \mathbb{R},$$

$$x \longmapsto y = f(x) = \frac{x+1}{2x-1}.$$

(1) We show that f is a well-defined function.

For every $x \in \mathbb{R} \setminus \{\frac{1}{2}\}$ the denominator $2x - 1 \neq 0$, hence the expression $\dfrac{x+1}{2x-1}$ is a real number. Therefore f assigns to each x in the domain a unique real value, so f is well-defined.

(2) We prove that f is injective. Is f bijective?

Let $x_1, x_2 \in \mathbb{R} \setminus \{\frac{1}{2}\}$ and we suppose that $f(x_1) = f(x_2)$. Then

$$\frac{x_1 + 1}{2x_1 - 1} = \frac{x_2 + 1}{2x_2 - 1}.$$

Cross-multiplying yields

$$(x_1 + 1)(2x_2 - 1) = (x_2 + 1)(2x_1 - 1).$$

Expanding and simplifying:

$$2x_1x_2 - x_1 + 2x_2 - 1 = 2x_1x_2 - x_2 + 2x_1 - 1$$

$$\Longrightarrow -x_1 + 2x_2 = -x_2 + 2x_1$$

$$\Longrightarrow x_1 = x_2.$$

Hence f is injective.

Next we check if f is surjective (onto) i.e. we check that $\forall y \in \mathbb{R}, \exists x \in \mathbb{R} \setminus \{\frac{1}{2}\} : y = f(x)$.

For every $y \in \mathbb{R}$, we solve $y = \dfrac{x+1}{2x-1}$. We obtain

$$y(2x-1) = x+1 \quad \Longrightarrow \quad 2xy - y = x + 1$$

$$\Longrightarrow x(2y-1) = y+1 \quad \Longrightarrow \quad x = \frac{y+1}{2y-1},$$

which is defined exactly when $2y - 1 \neq 0$, i.e. $y \neq \frac{1}{2}$. Thus no x is a preimage for $y = \frac{1}{2}$, so f is not surjective for the codomain $\mathbb{R}$. Therefore f is not bijective.

(3) The restriction that we should impose on the codomain so that f becomes a bijection is:

From the previous computation the range of f is $\mathbb{R} \setminus \{\frac{1}{2}\}$. If we take the codomain to be $\mathbb{R} \setminus \{\frac{1}{2}\}$, i.e.

$$f : \mathbb{R} \setminus \{\tfrac{1}{2}\} \longrightarrow \mathbb{R} \setminus \{\tfrac{1}{2}\},$$

then f is surjective (and we already showed it is injective), hence f is a bijection.

(4) We find the inverse function f^{-1}.

For the function $f : \mathbb{R} - \left\{\frac{1}{2}\right\} \longrightarrow \mathbb{R}$ is not bijective then f does not have an inverse function.

But $f : \mathbb{R} - \left\{\frac{1}{2}\right\} \longrightarrow \mathbb{R} - \left\{\frac{1}{2}\right\}$ is bijective and it has an inverse.

We already solved the equation $y = \dfrac{x+1}{2x-1}$ and we obtained

$$x = \frac{y+1}{2y-1},$$

x is unique for $y \in \mathbb{R} \setminus \{\frac{1}{2}\}$.

Therefore, the inverse function of $f : \mathbb{R} - \left\{\frac{1}{2}\right\} \longrightarrow \mathbb{R} - \left\{\frac{1}{2}\right\}$ is:

$$f^{-1} : \mathbb{R} - \left\{\frac{1}{2}\right\} \longrightarrow \mathbb{R} - \left\{\frac{1}{2}\right\},$$

$$y \longmapsto x = f^{-1}(y) = \frac{y+1}{2y-1}.$$

We observe that $f^{-1} = f$ $(f : \mathbb{R} - \left\{\frac{1}{2}\right\} \longrightarrow \mathbb{R} - \left\{\frac{1}{2}\right\})$

Solution exercise 16:

(1) An order relation must be reflexive, antisymmetric and transitive. We prove each property.

Reflexive.

What to prove: $\forall (x, y) \in \mathbb{R}^2,\ (x, y) \prec (x, y)$.

Proof: For any $(x, y) \in \mathbb{R}^2$ we have $x \leq x$ and $y \leq y$. Hence $(x, y) \prec (x, y)$. Therefore $\prec$ is reflexive.

Antisymmetric.
What to prove: $\forall (x, y), (x', y') \in \mathbb{R}^2,\ [(x, y) \prec (x', y')$ and $(x', y') \prec (x, y)] \Rightarrow (x, y) = (x', y')$.
Proof: Suppose $(x, y) \prec (x', y')$ and $(x', y') \prec (x, y)$. Then

$$x \leq x' \quad \text{and} \quad y \leq y',$$

and

$$x' \leq x \quad \text{and} \quad y' \leq y.$$

Combining $x \leq x'$ and $x' \leq x$ gives $x = x'$. Combining $y \leq y'$ and $y' \leq y$ gives $y = y'$. Thus $(x, y) = (x', y')$. Therefore $\prec$ is antisymmetric.

Transitive.
What to prove: $\forall (x, y), (x', y'), (x'', y'') \in \mathbb{R}^2,\ [(x, y) \prec (x', y')$ and $(x', y') \prec (x'', y'')] \Rightarrow (x, y) \prec (x'', y'')$.
Proof: Suppose $(x, y) \prec (x', y')$ and $(x', y') \prec (x'', y'')$. Then

$$x \leq x' \quad \text{and} \quad y \leq y',$$

and

$$x' \leq x'' \quad \text{and} \quad y' \leq y''.$$

Thus,

$$x \leq x' \leq x'' \quad \text{and} \quad y \leq y' \leq y''.$$

Hence $x \leq x''$ and $y \leq y''$, so $(x, y) \prec (x'', y'')$. Therefore $\prec$ is transitive.

Since $\prec$ is reflexive, antisymmetric and transitive, it is an *order relation* on $\mathbb{R}^2$.

(2) Total or partial?
We determine whether for every pair $(x, y), (x', y')$ we have either $(x, y) \prec (x', y')$ or $(x', y') \prec (x, y)$. If yes, the order is total; otherwise it is not.
The order is *not* total in general. For a counterexample we take

$$(x, y) = (1, 0), \qquad (x', y') = (0, 1).$$

We have $1 > 0$. Neither $(1, 0) \prec (0, 1)$ nor $(0, 1) \prec (1, 0)$ holds, so the order is partial.

Solution of exercise 17:
(1) An equivalence relation must be reflexive, symmetric and transitive. We prove each property.

Reflexive.
What to prove: $\forall A \in \mathcal{P}(E),\ A\mathcal{R}A$.
Proof: Let $A \in \mathcal{P}(E)$ we have $A = A$, so by the definition of $\mathcal{R}$ we get $A\mathcal{R}A$. Thus $\mathcal{R}$ is reflexive.

Symmetric.
What to prove: $\forall A, B \in \mathcal{P}(E),\ A\mathcal{R}B \Rightarrow B\mathcal{R}A$.
Proof: Let $A, B \in \mathcal{P}(E)$, we suppose $A\mathcal{R}B$. Then either $A = B$ or $A = \complement_E B$.

- If $A = B$, then $B = A$, so $B\mathcal{R}A$.
- If $A = \complement_E B$, then taking complements gives $B = \complement_E(\complement_E B) = \complement_E A$, so $B\mathcal{R}A$.

Hence $\mathcal{R}$ is symmetric.

Transitive.
What to prove: $\forall A, B, C \in \mathcal{P}(E),\ [A\mathcal{R}B \text{ and } B\mathcal{R}C] \Rightarrow A\mathcal{R}C$.
Proof: Let $A, B, C \in \mathcal{P}(E)$, we assume $A\mathcal{R}B$ and $B\mathcal{R}C$. There are four cases:

1. If $A = B$ and $B = C$, then $A = C$, so $A\mathcal{R}C$.
2. If $A = B$ and $B = \complement_E C$, then $A = \complement_E C$, so $A\mathcal{R}C$.
3. If $A = \complement_E B$ and $B = C$, then $A = \complement_E C$, so $A\mathcal{R}C$.
4. If $A = \complement_E B$ and $B = \complement_E C$, then $A = \complement_E \complement_E C = C$, so $A\mathcal{R}C$.

In every case $A\mathcal{R}C$ holds. Thus $\mathcal{R}$ is transitive.

Therefore $\mathcal{R}$ is reflexive, symmetric and transitive, so it is an equivalence relation on $\mathcal{P}(E)$.

(2) Equivalence class of a subset A.
For fixed $A \in \mathcal{P}(E)$, we determine $\overline{A} = \{B \in \mathcal{P}(E) : B\mathcal{R}A\}$.
By the definition of $\mathcal{R}$,

$$B\mathcal{R}A \iff B = A \text{ or } B = \complement_E A.$$

Hence the equivalence class of A is

$$\boxed{\overline{A} = \{\, A,\ \complement_E A \,\}}.$$

Solution of Exercise 18

1. (a) **False.** The odd integers are not closed under addition, we can take for example $3 \in H$ and $5 \in H$ such that $3 + 5 = 8$ and $8 \notin H$ because 8 is an even number. Or we can justify simply by $0 \notin H$ (0 the neutral element of $(\mathbb{Z}, +)$) because 0 is even. So $(H, +)$ is not a subgroup of $(\mathbb{Z}, +)$.

   (b) **True.** The set $[-1, +\infty[$ equipped with the addition is not a subgroup of $(\mathbb{R}, +)$. We can take for example: the inverse of $2 \in H$ is -2 and $-2 \notin H$.

   (c) **False.**

      i. $Id_E \in H$ $(H \neq \emptyset)$, the neutral element of G, is an element of H.

      ii. Let $f, g \in H$, we have
      $$f \circ g : E \mapsto E$$
      and f and g are bijective then, $f \circ g$ is bijective.
      hence, $f \circ g \in G$ (we can deduce this directly because G with $\circ$ is a group) .
      Also, we have
      $$f \circ g(x) = f(g(x)) = f(x) = x$$
      (because f and g are in H so $f(x) = x$ and $g(x) = x$).
      Thus, $f \circ g \in H$

      iii. Let $f \in H$, thus f is bijective then has an inverse function f^{-1} which is also bijective and
      $$f^{-1} : E \mapsto E,$$
      hence, $f^{-1} \in G$ (we can deduce this directly because G with $\circ$ is a group).
      Also, we have
      $$f(x) = x \Longrightarrow f^{-1}(f(x)) = f^{-1}(x) \Longrightarrow f^{-1}(x) = x$$
      (because $f \in H$ then $f(x) = x$).
      Thus, $f^{-1} \in H$

We conclude that $(H, \circ)$ is a subgroup of $(G, , \circ)$.

(d) **True.** we check the following subring conditions:

i. T **is nonempty.**
Since $0 = 3 \cdot 0$ and $0 \in \mathbb{Z}$, we have $0 \in T$, so $T \neq \varnothing$.

ii. We prove $\forall a, b \in T : \ a + (-b) \in T$
Let $a, b \in T$. Then there exist $m, n \in \mathbb{Z}$ such that $a = 3m$ and $b = 3n$. Hence

$$a + (-b) = 3m + (-3n) = 3(m - n).$$

Since $m + (-n) \in \mathbb{Z}$, it follows that $a + (-b) \in T$.
Remark:
(1) $-b$ is the inverse of b under addition.
(2) Proving that $\forall a, b \in T : \ a + (-b) \in T$ is equivalent to prove $(T, +)$ is a subgroup of $(\mathbb{Z}, +)$.

iii. We prove $\forall a, b \in T : \ a \cdot b \in A$

Let $a, b \in T$ with $a = 3m$ and $b = 3n$ for $m, n \in \mathbb{Z}$. Then

$$a \cdot b = (3m)(3n) = 9(mn) = 3(3mn).$$

Since $3mn \in \mathbb{Z}$, we have $a \cdot b \in T$.

From the previous properties, and using the proposition for characterizing a subring, we conclude $(T, +, \cdot)$ is a subring of $(\mathbb{Z}, +, \cdot)$.

2.
- The map $\varphi(x) = |x|$ is a group homomorphism . Indeed $G = (\mathbb{R}^*, \cdot), H = (\mathbb{R}^*, \cdot)$ are groups and we have $\varphi(x \cdot y) = |x| \cdot |y| = \varphi(x) \cdot \varphi(y)$.
Furtheremore, φ is not injective ($3 \neq -3$ and $|3| = |-3|$) then, it is not bijective, therefore, φ is not a group isomorphism .
- The map $\varphi(x) = 2x$ is not a group homomorphism . A group homomorphism must satisfy

$$\varphi(x \cdot y) = \varphi(x) \cdot \varphi(y).$$

We have

$$\varphi(xy) = 2xy, \qquad \varphi(x)\varphi(y) = (2x)(2y) = 4xy.$$

Since $2xy \neq 4xy$ in general, φ is not a group homomorphism then it is also not a group isomorphism.

Solution of Exercise 19

1) First part

1. We prove that $\perp$ is an internal composition law on G.
Let $x, y \in G$, we have $x \perp y = x + y - 1 \in \mathbb{R}$ (we have $x, y, -1 \in \mathbb{R}$ and the set of real numbers is closed under addition).

2. We prove that $(G, \perp)$ is an abelian group. We verify the group properties:

- Associativity of $\perp$:
Let $x, y, z \in G$, we have

$$(x \perp y) \perp z = (x + y - 1) + z - 1 = x + y + z - 2,$$

$$x \perp (y \perp z) = x + (y + z - 1) - 1 = x + y + z - 2.$$

Both sides are equal.

- Identity element:
  We prove that there exists $e \in G, \forall x \in G: \ x \perp e = e \perp x = x$.
  We have
  $$x + e - 1 = x \ \text{ and } e + x - 1 = x,$$
  both equations gives $e = 1 \in G$.
- Inverse element:
  We prove that $\forall x \in G, \exists y \in G: \ x \perp y = y \perp x = 1$. Let $x \in G$, we solve $x \perp y = 1$ and $y \perp x = 1$.
  Both equations gives
  $$y = 2 - x.$$
  and $y = 2 - x \in G$.
  Thus, the inverse of x is $2 - x$.
- Commutativity of $\perp$:
  We prove that $\forall x, y \in G: \ x \perp y = y \perp x$.
  Let $x, y \in G$, we have
  $$x \perp y = x + y - 1 = y + x - 1 = y \perp x.$$

From the previous properties we conclude that $(G, \perp)$ is an abelian group.

2) On the same set $G = \mathbb{R}$, we consider a second law $*$ defined by

$$\forall x, y \in G: \qquad x * y = xy - x - y + 2.$$

With the two operations $\perp$ and $*$ defined above on $G = \mathbb{R}$:

1. We prove that the structure $(G, \perp, *)$ is a ring.
   1) We already proved that $(G, \perp)$ is an abelian group.
   2) a) We prove that $*$ is internal composition law.
   Let $x, y \in G$, we have: $xy \in G$ ($\mathbb{R}$ is closed under multiplication), $-x \in G$ and $-y \in G$ (inverse of a real number under addition) and $2 \in G$. Because $\mathbb{R}$ is closed under addition then $x * y \in G$.

   b) We prove that $*$ is associative, i.e. $\forall x, y, z \in G; (x * y) * z = x * (y * z)$.

   $$(x*y)*z = (xy-x-y+2)*z = (xy-x-y+2)z-(xy-x-y+2)-z+2 = xyz-xy-xz+x+y+z-yz.$$

   $$x*(y*z) = x*(yz-y-z+2) = x(yz-y-z+2)-x-(yz-y-z+2)+2 = xyz-xy-xz+x+y+z-yz.$$

   then, we get the equality.
   3) Finally, we prove that $*$ distributes over $\perp$.
   i) Left-Distributivity:
   Let $x, y, z \in G$, we have

   $$x * (y \perp z) = x(y + z - 1) - x - (y + z - 1) + 2 = xy + xz - 2x - y - z + 3.$$

   On the other hand:

   $$(x * y) \perp (x * z) = (xy - x - y + 2) \perp (xz - x - z + 2)$$

$$= (xy - x - y + 2) + (xz - x - z + 2) - 1 = xy + xz - 2x - y - z + 3,$$

thus, $x * (y \perp z) = xy + xz - 2x - y - z + 3 = (x * y) \perp (x * z)$.
ii) Right-Distributivity:

Let $x, y, z \in G$, we have in the same way of the previous property.

$$(y \perp z) * x = (y * x) \perp (z * x)$$

From 1), 2) and 3) $(G, \perp, *)$ is a ring.

2. *Commutativity*: Let $x, y \in G$, we have

$$x * y = xy - x - y + 2 = yx - y - x + 2 = y * x.$$

So the ring is commutative.

3. We determine whether $(G, \perp, *)$ is a field. $(G, \perp, *)$ is a commutative ring and it is unital because we have

There exists an element e' is an identity if for all $x \in \mathbb{R}$ we have,

$$x * e' = x \qquad \text{and} \qquad e' * x = x.$$

We have

$$x * e' = xe' - x - e' + 2.$$

We solve $x * e' = x$, so

$$xe' - x - e' + 2 = x.$$

Rearranging gives

$$e'(x - 1) = 2(x - 1).$$

Since this must hold for all $x \neq 1$, we obtain

$$e' = 2 \in G.$$

The same reasoning for the other side:

$$e' * x = 2x - 2 - x + 2 = x.$$

for all $x \neq 1$, we obtain

$$e' = 2.$$

We check the special case $x = 1$:

$$1 * e' = 1e' - 1 - e' + 2 = 2 - 1 - 2 + 2 = e' * 1 = 1,$$

so the formula also works when $x = 1$.

Thus, the identity element is

$$\boxed{e = 2 \in G}.$$

Therefore, the commutative ring is also unital.
We have shown that $(G, \perp, *)$ is a commutative unital ring. The identity element with respect to $\perp$ is $0_G = 1$, and The identity element with respect to $*$ is $1_G = 2$.

$(G, \perp, *)$ be a field, if every nonzero element (any element except the identity element with respect to $\perp$) must have an inverse.
We prove $\forall x \in G - \{1\}, \exists y \in G: \ x * y = y * x = 2$.
We solve only $x * y = 2$ because $*$ is commutative:

$$xy - x - y + 2 = 2 \quad \Rightarrow \quad (x-1)(y-1) = 1. \quad \Rightarrow \quad y = 1 + \frac{1}{x-1}$$

For fixed $x \neq 1$, the equation has a unique real solution $y = 1 + \frac{1}{x-1} \in \mathbb{R}$. Hence, every nonzero element in G has an inverse element in G for $*$.

Therefore, $(G, \perp, *)$ is a field.

Solution of Exercise 20
We check the homomorphism with

$$f(x) = x - 1, \qquad a \oplus b = a + b - 1, \qquad a \odot b = ab - a - b + 2.$$

$(\mathbb{R}, \oplus, \odot)$ is a ring(proved in the previous exercise) and $(\mathbb{R}, +, \cdot)$ is also a ring.

1. We verify the two conditions:

   - For all $a, b \in \mathbb{R}: f(a \oplus b) = f(a) \ + \ f(b)$.
     Let $a, b \in \mathbb{R}$, we have

     $$f(a \oplus b) = (a + b - 1) - 1 = a + b - 2,$$

     and the other side gives

     $$f(a) + f(b) = (a - 1) + (b - 1) = a + b - 2.$$

     Thus, $f(a \oplus b) = f(a) + f(b)$.
   - For all $a, b \in \mathbb{R}: f(a \odot b) = f(a) \ \cdot \ f(b)$.
     Let $a, b \in \mathbb{R}$, we have

     $$f(a \odot b) = (ab - a - b + 2) - 1 = ab - a - b + 1.$$

     On the other hand:

     $$f(a) \cdot f(b) = (a - 1)(b - 1) = ab - a - b + 1.$$

     Both expressions are equal.

   Thus f is a ring homomorphism.

2. The identity element in $\mathbb{R}$ for $\odot$ is 2, while in $\mathbb{R}$ for $\cdot$ is 1. We check:

   $$f(2) = 2 - 1 = 1,$$

   f sends the identity element for the second law of the first ring to the identity element for the second law of the second ring and we have from the previous question that f is a ring homomorphism. Therefore f is a unital ring homomorphism.

7 Final Exam

Exercise 1:

Answer **True** or **False** for each of the following.

1. The algebraic structure $(\mathbb{N}, +)$ is a group. .....
2. For every proposition p, the proposition (p or $\bar{p}$) is always true. .....
3. Let $(G, \star)$ and $(H, \triangle)$ two groups, and h a function. If h is a group homomorphism from $(G, \star)$ to $(H, \triangle)$ then, $h(e_H) = e_G$. .....
4. For any non-empty set E, the binary relation $\subset$ represents a total order relation in $\mathcal{P}(E)$. .....
5. The function $g : \mathbb{R} \to \mathbb{R}$ defined as $g(x) = |x|$ is bijective. .....

Exercise 2:

Let the function f defined from $\mathbb{R}\backslash\{1\}$ to $\mathbb{R}$ by $f(x) = \frac{1}{x-1}$ and $\mathcal{R}$ be the binary relation defined on $\mathbb{R}\backslash\{1\}$ by

$$\forall x, y \in \mathbb{R}\backslash\{1\} : \quad x\mathcal{R}y \iff |f(x)| = |f(y)|$$

1. Determine $f^{-1}(\{1, -1\})$.
2. Prove that $\mathcal{R}$ is an equivalence relation.
3. Find the equivalence class of 0.

Exercise 3: Consider the set $G = \mathbb{R}\backslash\{-1\}$, and the law $\odot$ defined as

$$\forall x, y \in G : \qquad x \odot y = x + y + x.y$$

1. Prove that $\odot$ is an internal composition law in G.
2. Prove that $(G, \odot)$ is an abelian group.

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